What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network

Nikita L. Polomoshnov
Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University, Moscow, Russia
Institute of Biomedical Chemistry (IBMC), Moscow, Russia
nikitapol@fbb.msu.ru

August 29, 2026

Abstract

Cycle closure is the standard consistency check on a perturbation network, and what such a check can see is fixed by the gradient–cycle decomposition of edge flows (HodgeRank): the cycle part of a network's error is auditable and biases nothing, while the gradient part biases the affinities and is invisible to any statistic computed from the network's own values, at any magnitude. Both parts are measured here on 1145 real binding edges. The auditable subspace spans a third of the error space and, in the variance-weighted metric the closure statistic uses, carries under one per cent of the error against experiment. Experimental and annotation error is per-ligand, so it falls in the invisible part by construction and can only have made that share smaller: the measurement is a floor, and it survives correction: in the metric the closure statistic uses on every system measured, at one cell as a tie, and in an isotropic reading on every system of the wider grounded set, failing only on the two weakest of the four the test itself selected. The test built on it behaves as the bound predicts, and its flag set is not stable: the per-edge residuals and the ranking of systems reproduce across independent repeats, and the list of flagged systems does not. Fitting closure against the precisions the estimator already reports gives that test its stated null; it ties the standard fixed 1.0 kcal/mol cutoff, and on every replicate the set of systems it flags lies inside the set that cutoff flags. It flags nothing once the reported errors are widened by half, and published estimates of the pose and preparation component the protocol repeats hold fixed run above that, so the second ceiling sits lower than the first. It names no removal that recovers accuracy against experiment. Resolving the gradient part needs measurements from outside the network, at worst one per unresolved dimension.

1 Introduction

Relative binding free energy calculations are the most accurate computational tool available for ranking how tightly a set of candidate molecules will bind a target, and structure-based drug discovery programs increasingly build them into decisions about which molecules to synthesize. The calculations come as a network rather than as isolated numbers: each edge is an alchemical transformation between two related candidates, costing hours of molecular dynamics, and a realistic project touches dozens to hundreds of them. Because the affinities a program acts on are read off the network as a whole and not off any single edge, the network's own internal agreement is what a practitioner has to judge the calculations by before any molecule is made.

That judgement has a standard instrument. A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop in the network is zero in expectation, and how far a network fails to close is routinely reported as a diagnostic and acted on through fixed hysteresis cutoffs. The cycles meant here are loops over several edges of the network, the object of cycle-closure analysis; the *thermodynamic* cycle is the alchemical square behind a single edge, and the two are kept apart throughout. What has not been asked of the diagnostic is how much of a network's error it can reach at all: whether a network that closes its cycles is thereby a network whose affinities are right, and if not, what share of what is wrong with it a closure check could ever respond to.

Connectivity and redundancy answer different questions, and only the second is asked here. A connected network determines every $\Delta\Delta G$ up to one offset per component, redundant or not, which is all estimation needs. Detection is separate: it asks which errors leave a trace in the values the network reports, and the answer depends on redundancy rather than on connectivity.

That capacity follows from the algebra of the network rather than from any particular protocol, and it is measurable on public data. A network's per-edge error splits, uniquely and orthogonally, into a part carried by its cycles and a part that is a difference of per-ligand offsets. The first is the part any closure check sees, and it does not bias the estimated affinities at all; the second is the part that biases them, and no statistic computed from the network's own edge values responds to it at any magnitude. That fixes the capacity of a closure check; how much of a real network's error falls inside it is then a measurement. Both are made here, on a public benchmark of real binding edges with independent protocol repeats, and both are small: the auditable subspace spans a third of the error space, and the share of the error against experiment that falls inside it is under one per cent. Figure 1 runs the whole of that from left to right.

The test that runs against this bound is built from an uncertainty the calculation already supplies rather than from one learned alongside the prediction: where the prevailing approach fits a mean–variance head [23] or reads uncertainty off the disagreement of an ensemble [18], the estimator that turns raw simulation output into a free energy for a single edge [4] has a closed-form finite-sampling variance computed from that same output at no label cost. The bar the test carries is the multistate standard error that same calculation reports, of which that closed form is the two-state case, and validating it against independent repeats is what turns cycle closure from a descriptive check into a test with a stated null.

Section 2 places the work against prior free-energy estimators, learned uncertainty and cycle-closure diagnostics. Section 3 gives the decomposition and what it fixes about any closure-based check, and Section 4 measures where a real network's error lies against that bound. Section 5 characterises the per-edge error bar the test uses as its null and validates it against independent repeats, and Section 6 builds the test and reports what it inherits from the bound. Section 7 gives the data, the estimators, the models and the simulations behind every number reported above, and Section 8 collects the limitations, including an applicability map for the calibrated uncertainty downstream.

2 Related work

BAR [4] and MBAR [26] are the standard alchemical estimators, brought to prospective potency prediction by FEP+ [29], assessed at scale [22, 25], and carrying as their per-edge uncertainty the sandwich variance of M-estimation [12, 30]; the prevailing alternative learns that uncertainty from training examples instead [1, 15, 18, 23]. Cycle closure is long established as a network diagnostic [5, 29], as a weighting and estimation problem [7, 19, 31], and as necessary but not sufficient [22]. The gap addressed here is the quantitative form of that insufficiency: how much of a network's error a closure statistic can reach, and how much of a real network's error falls there.

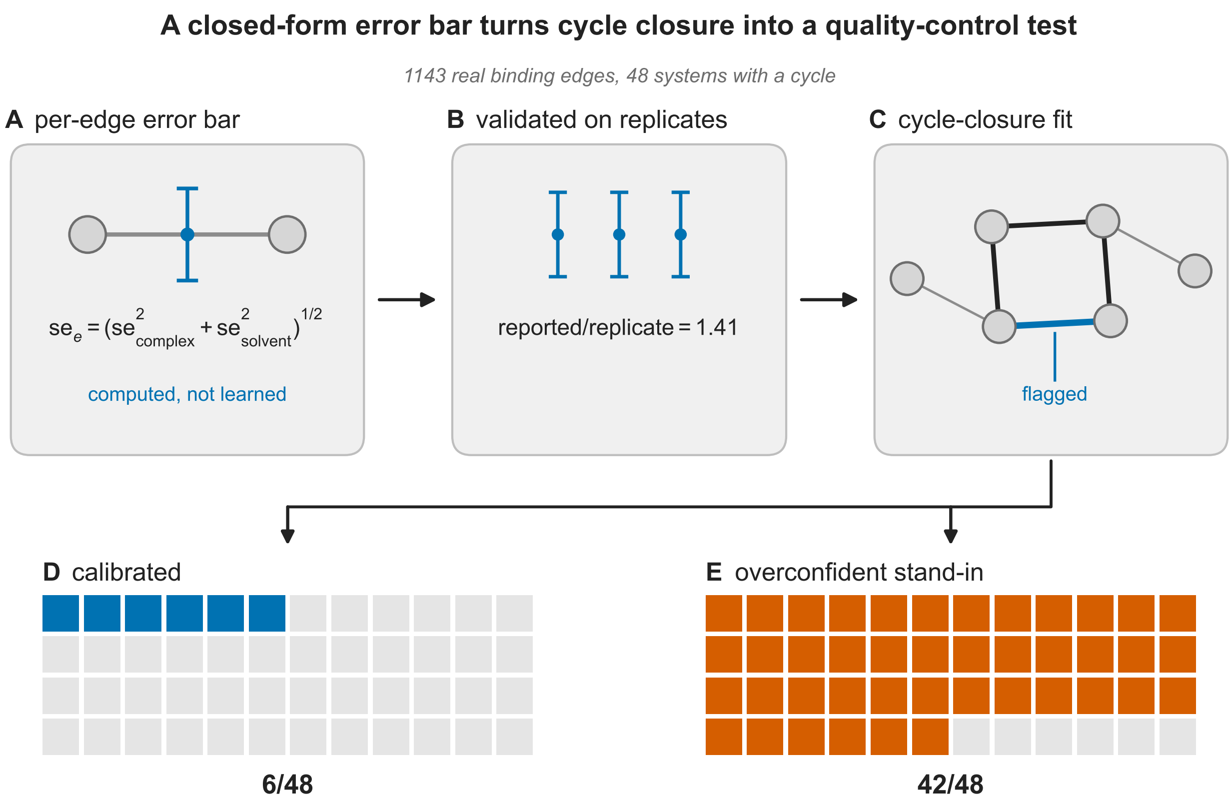

Figure 1: **Overview.** The pipeline this article runs on the benchmark, left to right. Every edge arrives with a per-edge standard error the estimator computes rather than learns, composed in quadrature from the complex-leg and solvent-leg uncertainties; that bar is validated against the benchmark's three independent protocol repeats (reported over replicate spread 1.41); and it is then the null of a generalized-least-squares cycle-closure fit, drawn as a small perturbation network with one cycle highlighted and one edge marked as flagged. Across the 1143 real binding edges that lie in the 48 benchmark systems carrying a cycle (of 1145 edges over 49 systems), that calibrated bar flags 6, 6 and 3 systems on the three repeats, one of them on all three, and a comparison bar built edge by edge from a learned head's measured miscalibration profile, not that head run here, flags 42.

3 What cycle closure can see: the decomposition

A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop is zero in expectation. Everything a closure check reports is a function of the residuals of that constraint, so what such a check can see is fixed before any protocol is chosen, by the algebra of the network alone. Fix the fit those residuals come from. Node potentials are estimated by generalized least squares on the signed incidence matrix, whitened by the reported per-edge standard errors, so that the conductances are $w_e = 1/V_e^{\text{rep}}$ with V_e^{rep} the per-edge variance the free-energy estimator reports (Section 5; Theorem S1 in the Supporting Information identifies that conductance with the sandwich precision I_e^2/B_e whenever the work samples are available, which on the benchmark used here they are not, so V_e^{rep} is the reported variance composed from the complex and the solvent leg, Section 7). The weighted sum of squared residuals of that fit is the closure statistic whose null Section 6 turns into a test; only its geometry is needed here. Two results follow, one per edge and one about the whole error space.

Theorem 1 (Estimation–detection conservation law)**.** *In the whitened GLS fit let $\Pi = \mathbb{I} - H$ be the residual-maker projector onto the cycle space (Π, to keep M for M-estimation and R for a Pearson correlation). Its diagonal, the curl-leverage $h_e = \Pi_{ee}$, satisfies $h_e + w_e\Omega_e = 1$ with $w_e = 1/V_e^{\text{rep}}$ and Ω_e the effective resistance of edge e [16]; hence $\sum_e h_e = \text{dof}$, and the closure-χ^2 noncentrality against a systematic shift μ on edge e is $h_e\mu^2/V_e^{\text{rep}}$. A bridge edge (in no cycle) has $h_e = 0$ and is undetectable at any μ; a node-consistent (gradient) bias lies in* $\text{range}(H)$ *and is annihilated by* Π*. A closure check's blindness to node-consistent force-field bias is therefore a* derived *property, not an assumption, and h_e is a per-edge observability certificate with resolution $\delta_e^* = \sqrt{V_e^{\text{rep}}/h_e}$, the shift at which the closure statistic carries unit noncentrality.*

The algebra is standard: $h_e = \Pi_{ee}$ is the hat-matrix leverage of the whitened design [11], its complement $w_e\Omega_e$ is the correspondence between leverage and effective resistance that graph sparsification uses [27], and $\sum_e h_e = \text{tr}(I - H) = \text{dof}$ is the trace identity for a projector. What the theorem adds is the reading of h_e as a per-edge observability certificate, which turns the test's blindness to node-consistent bias into a consequence rather than an assumption and names, before any simulation is run, the edges that can carry no evidence at all. The resulting map lists the structurally un-auditable (bridge) edges per system; on this benchmark the curl-leverage does not additionally predict the *location* of reproducible systematic error (studentized Spearman $\rho = -0.073$, permutation $p = 0.9571$, not significant; Supplementary Figure S9), so only the structural certificate is claimed.

Theorem 1 states the diagonal of a result about the whole error space, and that full result is what bounds the method. Write the per-edge systematic error as a vector μ over the E edges. Whitening and projecting splits it, orthogonally and uniquely, into a part that lies in the cycle space and a part that is a difference of per-ligand offsets.

Theorem 2 (Auditable and unauditable error)**.** *In the model $y = A\varphi + \mu + \varepsilon$ with A the signed incidence matrix of a network of N ligands, E edges and c components, and φ unknown:* (i) *for every δ, the parameter pairs (φ, μ) and $(\varphi + \delta, \mu - A\delta)$ induce the same distribution of y, so μ is identifiable only through $\Pi\tilde{\mu}$. The auditable subspace has dimension* $\text{dof} = E - N + c$ *and its orthogonal complement, the gradient fields $\mu_e = b_j - b_i$, has dimension $N - c$ and is unidentifiable: no statistic computed from a network's own edge values distinguishes a per-ligand bias from no bias, at any magnitude.* (ii) *The bias such an error induces in the fitted potentials, $(\tilde{A}^\top\tilde{A})^+\tilde{A}^\top\tilde{\mu}$, depends only on the gradient part, while the closure noncentrality $\|\Pi\tilde{\mu}\|^2$ depends only on the cycle part. The detectable part and the biasing part are exact orthogonal complements.* (iii) *For an error field with no* *preferential alignment, the expected visible fraction* $\|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ *is* dof/E. (iv) *Network estimation of* $\Delta\Delta G$ *under loop closure, with and without experimental constraints, is established [7, 31]; what the rank counting adds is the cost of the constraints. Given unbiased absolute measurements on a set* S *of ligands, the unidentifiable dimension is* $(N - |S|) - c_{\mathrm{free}}(S)$, *with* c_{free} *the number of components holding none of them: the first anchor in a component removes only that component's arbitrary offset and no unidentifiable direction, each further anchor removes exactly one dimension, and resolving a per-ligand bias in full requires an absolute measurement on every ligand.*

The decomposition is not new. Splitting an edge field on a graph into a gradient part and a divergence-free part, and reading the size of the second as whether the first is meaningful at all, is HodgeRank [13], where the gradient component is the global ranking being estimated and the cyclic component certifies it; a third, harmonic term appears there under a choice of two-cells and is not needed here. The hat-matrix and leverage machinery of part (ii) is likewise standard on networks of relative comparisons, where it already locates inconsistency in network meta-analysis [17]. That the closure check sees only cycles is in this sense a restatement, and it matches the long-standing reading of cycle closure as necessary and not sufficient [22].

The count in part (iii) is older still and closer to this setting than HodgeRank is. A network of relative comparisons under a consistency assumption has, in the language of mixed treatment comparisons, inconsistency degrees of freedom equal to the number of independent loops [21], which is dof here; the modern treatment of that model and its multi-arm complications is Higgins et al. [10].

Part (i) states that a bias common to every comparison involving one node is absorbed into that node's effect and is invisible to any inconsistency test. That literature does not state it in those terms, but it is immediate in that framework and may well be folklore there, as the corresponding fact is here, so the absence of an explicit statement does not establish that the fact is new.

The bias-adjustment strand of the same literature weakens the substance further. Bias-adjustment models in mixed treatment comparisons [6] treat study-level bias terms as quantities the network cannot identify on its own and inform them with external evidence or informative priors, which is part (i) for a different index and the same conclusion: a bias living where the design cannot separate it is resolved only from outside. The orthogonality of part (ii) and the anchor counting of part (iv) are likewise not located in that strand, and not finding them there does not establish that they are absent.

Part (ii) follows from part (i) in two lines. The bias in the fitted potentials is $(\tilde{A}^\top\tilde{A})^+\tilde{A}^\top\tilde{\mu}$ and $\tilde{A}^\top\Pi = 0$, so the gradient part alone determines it while $\|\Pi\tilde{\mu}\|^2$ alone determines the noncentrality; it is the defining property of an orthogonal projector, read on the error rather than on the signal. What is left to this article is that reading for alchemical networks and the measurement of where a real network's error falls between the two halves, which is Section 4 and what the rest of this article rests on. Part (iv) is a dimension count and is not asserted as new. Network estimation of $\Delta\Delta G$ under loop closure is solved in Xu [31] by Fisher information and in Giese and York [7] by a constrained variational solution of the whole network; neither is the object part (iv) counts, which is the number of absolute measurements needed to remove the unidentifiable directions. That counting may well be present in either, and it is used here to state that cost.

The map as a design rule. The observability certificate is prospective: alone among the quantities measured here it can be acted on before a simulation is run. It prices two distinct purchases on the replicate benchmark (1143 edges over the 48 systems that carry a cycle; Section 7). Coverage costs 29 added ligand pairs, 2.5% more perturbations than were already run, and leaves no bridge anywhere; connecting every system as well costs 36. Both counts are certified minimum against the Eswaran–Tarjan augmentation bound and assume nothing about the edges they add (Supplementary Table S5; `make figDesign`).

Resolution is the second purchase, and the certificate prices it together with the point at which it stops being purchasable. An edge freshly placed on one long cycle has small h_e and correspondingly poor δ_e^*, so coverage does not buy resolution and the two are costed apart: where a target is reachable and not already met, a greedy design reaches a median δ^* of 1.0 kcal/mol on 39 of the 48 systems for a median of two further edges, on an assumed variance for edges not yet run. Past that point the graph is no longer the binding constraint. Because $h_e \leq 1$ pointwise, no design pushes an edge's resolution below its own reported standard error, and measuring every remaining ligand pair still leaves nine systems with a median δ^* above 1.0 kcal/mol and nineteen above 0.5. There the sampling budget is what has to change, which is the bound of Section 3 presenting itself as a design constraint rather than as a measurement.

Two properties of the certificate carry into the rule. Its targets are on a resolution scale and not a detection threshold, at the ratio Section 4 measures; and it says where the test could see, not where reproducible error lands, which the same curl-leverage does not predict.

4 Where the error actually lies

Theorem 2 makes two quantities measurable that were previously matters of assumption: how much of a network is auditable at all, and how much of its real error the audit can reach. Both are answered on this benchmark in Figure 2.

Of the 1143 edge dimensions carried by the 48 systems with a cycle, 372 lie in the cycle space, so two-thirds of the error space is node-consistent and invisible to every closure-based diagnostic, this one included. Forty-eight edges are bridges, carrying no evidence at any magnitude. Half the remaining edges cannot resolve a shift below 0.79 kcal/mol, and only 27% of all 1143 can resolve one of 0.5 (Figure 2A). Neither figure is a detection threshold: requiring 80% power at $\alpha = 0.05$ moves that median to 2.2–3.7 kcal/mol.

That is the first of two reasons both numbers overstate the test's reach, and the two compose. δ^* is the shift at unit noncentrality: power there runs from 0.170 to 0.070 across the degrees of freedom present, and 80% at $\alpha = 0.05$ needs 2.8 to 4.7 times it. δ^* is also carried on the raw reported standard errors, where the calibrated scale is 0.92 of them (Section 6).

Composing the two gives operational numbers worse than δ^* alone suggests. Only 6.6% of auditable edges reach 80% power against a 1.0 kcal/mol shift on that edge alone, at $\alpha = 0.05$ and before any multiplicity correction. At the level the test actually fires, a system's most observable edge needs a median 1.35 kcal/mol for 80% power (quartiles 0.81 and 2.06) and its median edge needs 3.33; a third of systems could detect a 1.0 kcal/mol error somewhere, and 58% could detect 1.5. The regime the test operates in therefore begins at about the top of the band that relative-binding-free-energy decisions turn on, which is a sharper statement of the same limitation the rest of this section measures.

That is the capacity of the audit; its reach on real error is smaller, and measured on different graphs. Figure 2B carries two grounded readings of that reach on one shared axis: B1 the four sub-networks the closure test itself selected, on a ChEMBL join, and B2 the eight the benchmark's own curated per-edge labels reach, six of which the test does not flag. They run on different edges and different labels, neither corrects the other, and the differences between them are set out with the second reading below. Affinities cover part of each ligand set, so Figure 2B1 runs on each system's grounded sub-network rather than on the full networks of Supplementary Table S4. As (N, E, c, dof) these are $(24, 31, 2, 9)$, $(41, 59, 1, 19)$, $(34, 51, 1, 18)$ and $(23, 26, 2, 5)$ for `cdk8`, `hif2a`,

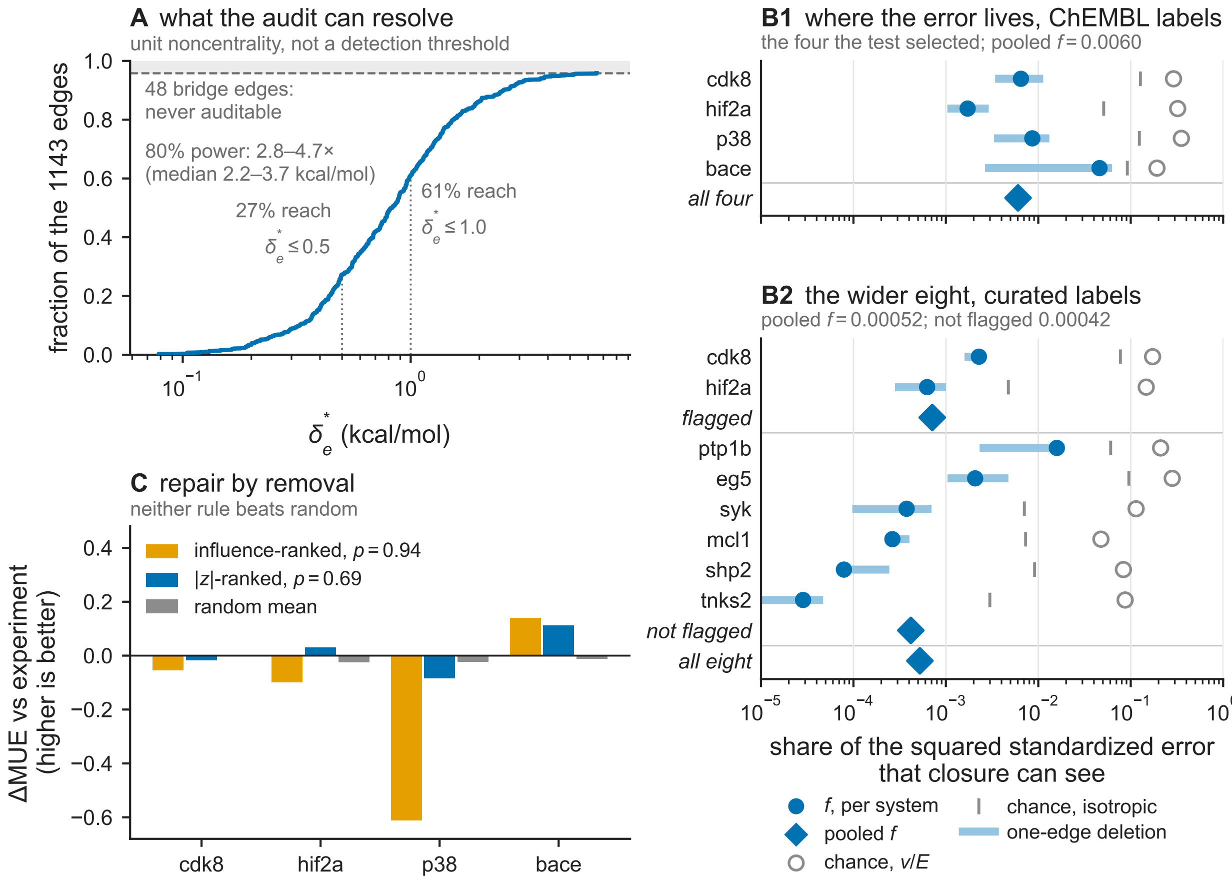

Figure 2: **Auditable capacity and the location of the error.** (A) Cumulative fraction of the 1143 benchmark edges against δ^*_e, the shift at unit noncentrality, with the bridge edges marked as the ceiling they impose. (B) Two grounded measurements of the same quantity, drawn on one shared axis so that they can be read against each other: the fraction f of a system's error against experiment lying in the auditable subspace, the fraction an unaligned error field would show under two conventions – whitened-isotropic (dof$/E$) and isotropic in kcal/mol – and the range f takes when one edge is deleted at a time. (B1) The four sub-networks the closure test itself selected, grounded on a stereo-blind ChEMBL join: 167 edges, 51 residual degrees of freedom. (B2) The eight systems whose curated per-edge labels the benchmark carries and whose matched sub-network keeps a cycle: 205 edges, 30 residual degrees of freedom, grouped by whether the test flags them, and the six it does not flag show the effect at least as strongly as the two it does. Neither block corrects the other and the two are not nested: `bace` is absent from the curated benchmark entirely, `p38`'s matched sub-network is acyclic, the edge sets and the labels differ, and B2 carries the fewer residual degrees of freedom. On edges held genuinely identical the curated labels give the *higher* f, which is the direction a per-ligand label error predicts. Italic rows are pooled and carry f alone; the chance levels each pooled figure is measured against are given in the text. (C) Change in mean unsigned error against experiment after removing the same number of edges chosen by influence, by standardized residual, and at random.

p38 and bace, with auditable fractions 0.290, 0.322, 0.353 and 0.192 rather than the tabulated ones (cdk8 is 0.46 on its full network); two are disconnected, which the fit absorbs into one offset per component.

All four sub-networks were selected by the calibrated closure test of Section 6, so what follows characterises the networks that test flags and not the benchmark. One of the four, cdk8, carries its flag from a naming collision between two releases rather than from either release's simulations.

The quantity measured on them is the visible fraction $f = \|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ of Theorem 2(iii), here the share of the measured error against experiment that lies in the auditable subspace. Pooled it is 0.6%, against those chance levels (Figure 2B1); per system 0.0065, 0.0017, 0.0086 and 0.0460. Dropping cdk8 leaves the pooled figure at 0.6% (0.0060 against 0.0060) and moves it from 5.45% to 6.10% in the unweighted metric, so the pooled figure does not rest on it.

The per-system figures are less stable than the pooled one. Leaving out one edge at a time moves them by 1.9, 1.7 and 2.6 times on the first three and by 17 on bace, whose range spans a factor of 23.

Both sides of the ratio are measured in the variance-weighted metric the closure statistic lives in, and the margin over chance depends on that choice. It narrows to a factor of two on bace under isotropy in kcal/mol, and the pooled fraction rises to 5.45% in the unweighted metric, which recomputes the projector without variance weighting. Supplementary Table S2 gives all three conventions for all four systems with their own chance levels and their single-edge-deletion ranges; its weakest cell is bace isotropic under deletion, where f is smaller than its own chance level by a factor of 1.36.

Against the frozen H1 criterion, a pooled degrees-of-freedom-adjusted gradient-model R^2 of at least 0.50 and positive on three of the four systems, the pooled value is 0.980 and the per-system values are 0.978, 0.995, 0.976 and 0.761, so the criterion is met. Pooling is over the four sub-networks jointly, f as a ratio of summed squared norms and the adjustment carried on the summed $E = 167$ edges and rank 116, that is on 51 residual degrees of freedom, and not as an average of per-system adjustments.

The criterion evaluated is the degrees-of-freedom-adjusted rule the pre-registration froze and not the weaker unadjusted form $1 - f_{\text{pooled}} \geq 0.50$. These figures support a bound and not a point estimate: the visible fraction is small, and at its weakest it sits a factor of two below its chance level, not the two orders of magnitude the pooled figure alone suggests.

The measurement is not an artefact of the fit, because it is a near-identity: per-ligand affinities make the experimental $\Delta\Delta G$ vector a gradient field, so $\Pi\varepsilon$ reduces to the closure residual and f is the closure statistic over the total squared standardized error. It survives the obvious contaminants, holding on medians rather than sums and holding on bace, whose ligands carry no enantiomer pair collapsed by the stereo-blind join (Supporting Information). Changing the label source to the curated per-edge affinities the OpenFF protein–ligand benchmark of Section 7 carries, available for three of the four systems, agrees where it can be run (Supporting Information). Per-edge error against experiment runs 1.06–1.45 kcal/mol at the median against reported standard errors of 0.14–0.40: these networks are wrong by several standard errors while closing their cycles, in the one direction closure cannot look.

The design does not license attribution. Experimental measurement error and the annotation join are per-ligand, hence gradient fields too, so what is established is that the dominant part of the error is node-consistent, not that it is force-field bias.

The curated cross-check named above is not confined to the four systems the test selected, and it is the second block of the same panel. That source shares fourteen system names with the replicate set, and eight of them keep a cycle once the two edge sets are matched by structure rather than by name; pooled over those eight, on 205 edges and 31 residual degrees of freedom, the visible fraction is 0.00052 against chance levels of 0.151 whitened and 0.0137 isotropic (Figure 2B2; Supplementary

Figure S6 and Table S3; `make figGround`). The systems the closure test does not flag show the effect at least as strongly as the ones it does, 0.00042 over six against 0.00071 over two, which turns the measurement from a statement about test-selected networks into a statement about the benchmark.

The expanded measurement neither replaces the four-system one nor corrects it, which is why Figure 2B keeps the two as separate blocks rather than as one series. `bace` is absent from the curated set entirely, `p38`'s matched sub-network is acyclic and contributes to neither side, the two run on different edge sets, and the expanded set carries 31 residual degrees of freedom against the original's 51: the two are measurements of different things on different labels, and both are reported. On genuinely identical edges the cleaner labels give the *higher* f, by 4.4× on `cdk8` and 2.9× on `hif2a`. That is the direction a per-ligand label error predicts and not its reverse, because such an error is a gradient field: the whitened projector annihilates it, so it lands wholly in the denominator of f and pushes the noisier source's reading down.

One degree of freedom in the protocol was undeclared; the naming finding of Section 6 settles it. A ligand name is not an identifier here, so the figures quoted throughout key nodes by resolved structure, which is the less favourable reading: merging two names for one molecule knits components together and adds cycles, and cycles can only raise the auditable share. `ptp1b` is the only system of the eight it touches, going from $f = 0.0023$ to 0.0159, its worst single-edge-deletion isotropic margin from 10.3× to 2.0×, and the pooled figure from 0.00046 to the 0.00052 above. Neither key is safe in general, and the Supporting Information shows where this one fails: a 2D structure merges poses a release names apart. Both readings are reported.

The label source can move f in one direction only, which is what makes it a bound. Those per-ligand errors reach an edge as a difference of node offsets, exactly the gradient fields part (i) describes, and the whitened projector annihilates them. Injecting a per-ligand field at the mixed-laboratory reproducibility public affinity data show, 0.68 $\log_{10}$ units per ligand [14], moves the numerator of f by at most 1.96×10^{-11} of itself over 800 seeded draws, against a norm of order one, while the denominator grows by 1.6 to 2.2 times at the median, which is the inflation the injection produces (Supplementary Figure S7; `make figNoise`). Label noise therefore contributes nothing to the numerator and all of it to the denominator, so a correction for it can only raise f, and the measured 0.6% is a floor on the share attributable to the calculation's own error and not an estimate of it.

The room that leaves is not uniform across systems, and the pooled crossing is not the reading to quote alone. Pooled, f would first reach its chance level at a denominator shrinkage of 50.6× whitened and 14.1× isotropic. What a correction could deliver is the reverse accounting: removing the share a literature-scale per-ligand noise implies would shrink the denominator by 4.13× pooled, once each system is held to its own arithmetic. `bace` needs only 4.2× whitened and 2.0× isotropic, and under the least favourable single-edge deletion `p38` needs 1.46× and `bace` 1.36× isotropic, which a plausible correction does supply.

The two metrics accordingly give different verdicts, and both are reported. The whitened reading, the metric f is defined in, survives every level tried on every system with one cell excepted. That cell is `bace` under the least favourable single-edge deletion, where the 2.55× needed and the 2.55× supplied by the level carrying half the mixed-laboratory per-edge variance coincide to within one per cent, which is a tie rather than a margin. The isotropic reading does not survive on `p38` and `bace`, the two weakest cells of the isotropic convention already tabulated.

How far a correction can be pushed follows from arithmetic. At the full mixed-laboratory level the implied label-noise share exceeds one on `cdk8`, `p38` and `bace`, asserting more label variance than those networks' entire measured error against experiment contains, so that level is an upper bound and not an estimate for a join that takes one assay per system and never mixes IC_{50} with K_i.

On `eg5`, whose ligands carry ChEMBL identifiers, the network $\Delta\Delta G$ from the same generalized-least-squares fit agrees with real KIF11 experimental affinity [33] at a mean unsigned error (MUE) of 0.79 kcal/mol (95% bootstrap CI $[0.59, 0.97]$, conditional on the fitted network; Pearson $R = 0.65$ $[0.30, 0.83]$, $p{\approx}2{\times}10^{-4}$, $n = 28$), and its clean cycle-closure reduced $\chi^2 = 0.64$ is consistent. That is not evidence that a clean system is accurate, and the five grounded systems cannot settle the question in either direction.

The ordering by mean unsigned error is `eg5` 0.79 = `p38` 0.79 < `bace` 0.88 < `cdk8` 0.91 < `hif2a` 1.47, so the one unflagged system ties the best flagged one and is below the other three. That direction favours the flag, but it rests on a single unflagged system against four, and the grounding is not matched: `eg5` comes from a single named ChEMBL assay while the four flagged systems come from a target-wide connectivity search under the coverage rule. No separation claim follows from it in either direction.

Acting on the flag does not improve accuracy on `eg5` (internal $|z|$ against error-against-experiment, $r = +0.13$), which is what the scope predicts: on a clean system the residual error against experiment is dominated by node-consistent force-field bias, and cycle closure cannot see that. The complementary test, that repairing a flagged system improves its accuracy, needs experimental affinities for the flagged systems; these are not directly tabulated in the replicate set but are recoverable by per-paper ChEMBL structure-search. Of the six flagged systems, four clear a 60% public-ChEMBL coverage gate (**`cdk8, hif2a, p38, bace`**; connectivity-based structure matching, one assay per system, IC_{50} before K_i, never mixed); their baseline network MUE against this grounded experimental affinity is 0.79–1.47 kcal/mol, typical-magnitude FEP error, which confirms the grounding itself is sound.

On these four, guided removal of the flagged high-$|z|$ edges (the same repair order as Figure 3A, fixing a removal count $K = 1$–3 per system) does not reduce MUE-against-experiment more than an equal-size random removal does (Supplementary Figure S10; per-system one-sided permutation $p = 0.905, 0.136, 0.880, 0.071$; combined Stouffer $p = 0.48$, sign test $p = 0.69$). Two systems improve slightly over their random-null mean and two worsen slightly, with no system individually significant. The frozen criterion is conjunctive, requiring Stouffer $p < 0.05$ and a guided ΔMUE, defined throughout as MUE(full) − MUE(after) so that an improvement is positive, above the 95th percentile of its own random null in a majority of grounded systems; neither conjunct is met. The remaining two flagged systems (**`brd4, faah`**) did not clear the coverage gate and are not included.

That null is what Theorem 2 predicts and what the paragraph above measures. A removal rule reads the visible part of the error; the visible part carries under one per cent of what is wrong with these networks, and by part (ii) it is the part that biases the fitted potentials least. A second pre-registered rule tests whether the ranking was simply the wrong one, and fails to find that it was; a second failure to reject is weaker evidence than a rejection would be. Part (ii) suggests ranking edges by influence rather than by residual. The statistic used is $|z_e|\sqrt{(1-h_e)/h_e}$ with $z_e = r_e/\sqrt{V_e^{\mathrm{rep}}}$ the raw standardized residual, which is the classical influence form scaled by $\sqrt{h_e}$ and not that form itself; edges with h_e below 10^{-9} are treated as bridges and given influence zero. Writing it in the unscaled form changes the removal set on two of the four systems, and that variant is reported with the outcome below.

At the same removal counts the influence rule performs worse than the rule it was meant to replace, against the same conjunctive criterion: better on one system of four (per-system ΔMUE -0.055, -0.099, -0.612, $+0.140$ kcal/mol at one-sided permutation $p = 0.937$, 0.819, 1.000, 0.023; Figure 2C), at a pre-registered sign test of $p = 0.94$ against the residual rule's 0.69. On four systems that test attains only $\{1.0000, 0.9375, 0.6875, 0.3125, 0.0625\}$, so neither value could reach its own level and the two do not form an ordering, one system flipping separating them.

The pre-registration's other combination, a one-sided Stouffer, returns no usable number here.

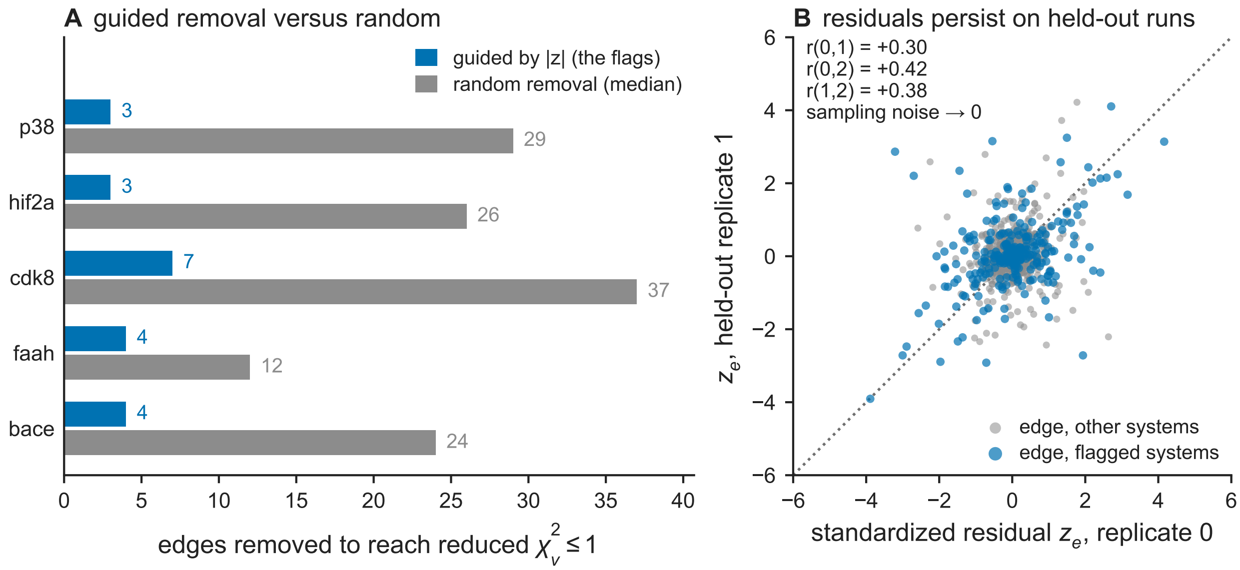

Figure 3: **Edge-removal repair test and cross-replicate residual reproduction.** (A) Number of edges removed, in descending order of $|z|$, before reduced $\chi^2 \leq 1$ is restored: guided removal of the flagged high-$|z|$ edges versus random removal, per system. (B) Standardized residual z_e from a replicate-0 network fit versus z_e from a replicate-1 fit, for edges in flagged and in non-flagged systems, with the $y = x$ line and the Pearson correlation between each pair of replicates.

One system's permutation p saturates at the resolution of its own grid and Stouffer maps that to an infinite z; clipping to the grid's range gives 0.96, and the residual rule, which has no saturated input, stands at 0.48.

Neither rule beats random removal, and on one system the influence rule does measurable harm. On `p38` its guided ΔMUE of -0.612 kcal/mol falls below every one of the 1000 random draws (minimum -0.416, no ties), so the opposite tail is at most 1/1001. That tail is post hoc, the pre-registration having asked whether the rule helps; it is one system of four, about 0.004 under a Bonferroni allowance; and the magnitude is convention-dependent, -0.612 under greedy removal that refits after each edge and -0.232 when the statistic is applied once to the original fit. The sign survives both conventions, and survives the strict influence form, which changes the removal set on two of the four systems but not here. The two edges removed carry $h_e = 0.099$ and 0.388, so the harm is not the degenerate branch where a near-bridge is divided by its own vanishing leverage.

The other tail points the other way and is reported alongside it. On `bace` the same rule improves the error against experiment by 0.140 kcal/mol at a one-sided $p = 0.023$, which the conjunctive criterion does not let the summary report, and which is exactly as post hoc and as single-system as the harm. Neither tail clears the pre-registered bar, and both are reported.

The rule fails overall for the reason measured above: the influence form divides a residual by a small curl-leverage to recover the error of a single anomalous edge, and a smooth per-ligand field is not concentrated on any edge, so the division returns amplified noise.

A flag remains a system-level signal that a protocol has produced results inconsistent beyond sampling, and that signal reproduces across replicates in its per-edge residuals and in its ranking of systems, though not in the list of names it returns. It is not a repair target: deleting the edges it names cannot recover accuracy against experiment, which is set by a component the audit is blind to by construction.

Reaching that component needs information from outside the network, and it has to be information the network does not already contain: an absolute binding free energy calculation carries exactly the force-field component of the per-ligand systematic error part (i) is about, so it does not resolve that component at all; the mapping, soft-core, dummy-atom, schedule and restraint contributions are protocol-specific and would not transfer. A measured anchor resolves its dimension only to its own precision. By part (iv) the requirement does not soften when the anchors are unbiased: one anchor per component fixes only an arbitrary offset, and every dimension after that costs one more measured ligand.

5 The per-edge error bar as a null hypothesis

The test of Section 6 needs a per-edge variance it can treat as known. Two quantities are involved, and they are kept apart throughout: the closed form for a single BAR window, which the replicate benchmark cannot exercise because it releases no work samples, and the bar the test actually carries, which is the multistate standard error that benchmark ships, composed in quadrature from the two legs and validated against independent repeats. Everything below consumes the variance as a number or as a fixed graph weight.

Theorem 3 (Sampling variance via the sandwich)**.** *For one alchemical edge with n_f forward and n_r reverse works, let p be the logistic BAR weights at the root $\widehat{\Delta F}$ of Bennett's estimating equation $S(\Delta F) = 0$ and $I = \sum p(1-p)$ its curvature, the work-distribution overlap. Then $\widehat{\Delta F}$ is an M-estimator, so its asymptotic sampling variance is the sandwich* $\mathrm{Var}(\widehat{\Delta F}) = A^{-1}BA^{-1} = B/I^2$*, with $A = I$ and $B = n_f \,\mathrm{Var}_f[p] + n_r \,\mathrm{Var}_r[p]$ [12]. Here $A \neq B$ arises from BAR's* stratified two-sample design *and not from model misspecification: the information equality $A = B$, which would give $1/I$, holds for a single-sample MLE and fails for a two-sample one. Evaluated at the root, with works drawn from the correct pair of ensembles, $B/I^2 = 1/I - (1/n_f + 1/n_r)$ identically: the sandwich is not an approximation to the Bennett/Shirts–Chodera (MBAR) uncertainty [4, 26] that* `pymbar` *reports but the same quantity, and $-(1/n_f + 1/n_r)$ is exactly the $O(1/n)$ gap between it and the naive plug-in $1/I$. The two* sample *plug-ins still differ through the moments they substitute, by a paired discrepancy falling monotonically and by more than an order of magnitude from n=40 to n=320, the range over which it exceeds Monte-Carlo noise at a modest replicate budget.*

Remark (scope of "calibrated"). B/I^2 is the variance of the *sampling* error of $\widehat{\Delta F}$, conditional on the works being drawn from the correct alchemical ensembles. It says nothing about systematic force-field bias, which is outside B/I^2; a separate epistemic correction term is attached where the downstream uses of the uncertainty are audited (Section 8). It is also non-negative by construction, where the plug-in expression can return a negative value at very high overlap with tiny samples.

Figure 4 checks the closed form on real work samples, the 19 adjacent-λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation. The sandwich over decorrelated-bootstrap truth is 1.00 $[0.96, 1.07]$, against 2–12× for naive $1/I$ on the same reference, every window sitting at overlap ≥ 0.84 (per-window values, Supplementary Table S1). That reference reuses the same work samples, so the figure establishes agreement with a decorrelated bootstrap on real molecular dynamics in the high-overlap regime this schedule occupies, and not a ligand-to-ligand binding quantity; and since the replicate benchmark releases no work samples, the work-sample route to B/I^2 is exercised on these 19 windows alone. Both the sandwich inputs and the bootstrap truth are subsampled to the statistical inefficiency g [26] before B is formed, so Figure 4 uses effective n (Supplementary Figure S1).

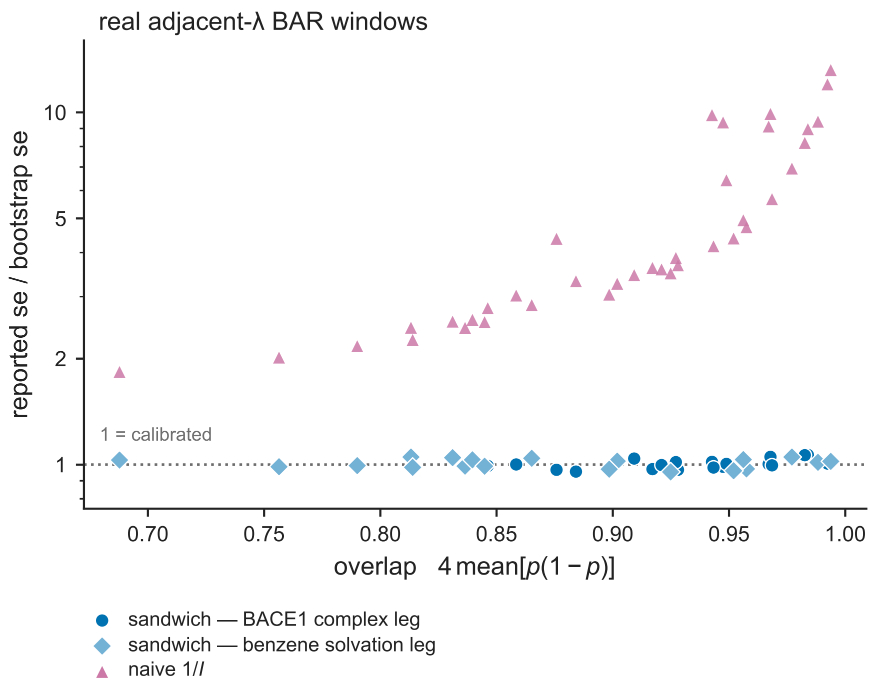

Figure 4: **Sandwich variance against a decorrelated bootstrap on real BAR windows.** Reported standard error divided by the decorrelated-bootstrap standard error, versus work-distribution overlap, for the sandwich B/I^2 on the 19 adjacent-λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation (filled circles) and on the 19 windows of the benzene solvation legs (filled diamonds), and for the naive information-equality plug-in $1/I$ on the same windows (triangles). Each point is one two-state BAR estimate between neighbouring λ values, and every window sits at overlap ≥ 0.84. The controlled low-overlap sweep and the learned-variance baselines are Supplementary Figure S2.

The low-overlap behaviour rests instead on a controlled Monte-Carlo overlap sweep, in which the closed form tracks MBAR and Monte-Carlo truth across the range at a reported/true se of ≈ 1 and at no Monte-Carlo cost, bare $1/I$ overstates by 1.08–2.65×, worst at high overlap, entering as the information-equality plug-in the sandwich corrects and not as a baseline anyone reports, and a learned mean–variance head given the same overlap information is overconfident at both label budgets tested; at extreme low overlap (n=20) the bias-corrected plug-in itself runs ∼10% below MBAR, both ≈ 1. That last result is a property of the head's per-edge parameterisation and not a limit on learning the quantity: pooling the same single labels by overlap recovers the sandwich to ∼6%, and a null miscalibrated by that much leaves the test of Section 6 intact. The sweep, and the two restrictions that travel with the learned head, are Supplementary Figure S2.

The reported bar against independent replicates. Figure 5 validates the bar the test actually uses, on samples it does not reuse. Each edge's reported bar is composed in quadrature from the complex-leg and solvent-leg MBAR uncertainties shipped with the benchmark, $\mathrm{se} = \sqrt{\mathrm{se}^2_{\text{complex}} + \mathrm{se}^2_{\text{solvent}}}$. Each leg term is a multistate MBAR uncertainty over that leg's λ windows and not the two-state BAR quantity of Theorem 3, so what is validated here is the bar as a practitioner receives it and not the two-state identity itself.

Across 1145 real binding edges from the OpenFE IndustryBenchmarks2024 [2] (3 independent replicates each), the reported MBAR se is calibrated-to-conservative against the across-replicate truth: the replicate ratio is 1.41 $[1.26, 1.62]$, and the bar is conservative on 72% of edges and on 28 of the 34 systems carrying at least eight edges. Both sides of that ratio pool in quadrature, so the n=3 replicate term is unbiased at the variance level and needs no small-sample correction. The three replicates are independent stochastic repeats of the sampling process from identical starting coordinates (OpenFE `protocol_repeats` [2]), so the across-replicate SD reflects sampling and convergence noise alone and is a *lower bound* on full reproducibility, excluding pose and system-preparation variance. The reported se does not systematically under-state real reproducibility.

That conclusion disagrees with the closest measurement in the literature. Wade et al. [28], comparing the analytic MBAR uncertainty against the spread over five independent replicas on 54 protein-ligand transformations across five targets in two engines, report that MBAR consistently under-estimates the error, by up to 0.9 kcal/mol. Two differences bear on the disagreement, in opposite directions. Their replicas resample the full ensemble, whereas the OpenFE repeats share starting coordinates, so the denominator used here is a lower bound on run-to-run variability and the ratio an upper bound on conservatism. Their own functional, applied to these edges, sharpens the disagreement instead of resolving it: the analytic standard error falls below the replicate spread on 0.279 $[0.224, 0.332]$ of edges, against the 0.368 correctly scaled bars produce at three replicates, and the interval excludes that value. Settling the question needs replicates that resample preparation and pose, which this benchmark does not provide, and its eight multi-release systems supply no usable substitute (Section 6).

6 The test, and what it inherits from the bound

The calibrated closure test, called the quality-control (QC) test in this article's figures and the calibrated test below, makes the bound operational: the residuals of the same projector whose diagonal Theorem 1 reads as an observability certificate, weighted by the bar of Section 5, are a statistic with a stated null. It separates the reproducible part of a perturbation network's edge-level inconsistency from its sampling noise. Candidate physical sources of a reproducible part include non-convergence, hysteresis, and charge, water or protonation artifacts, but this design does not

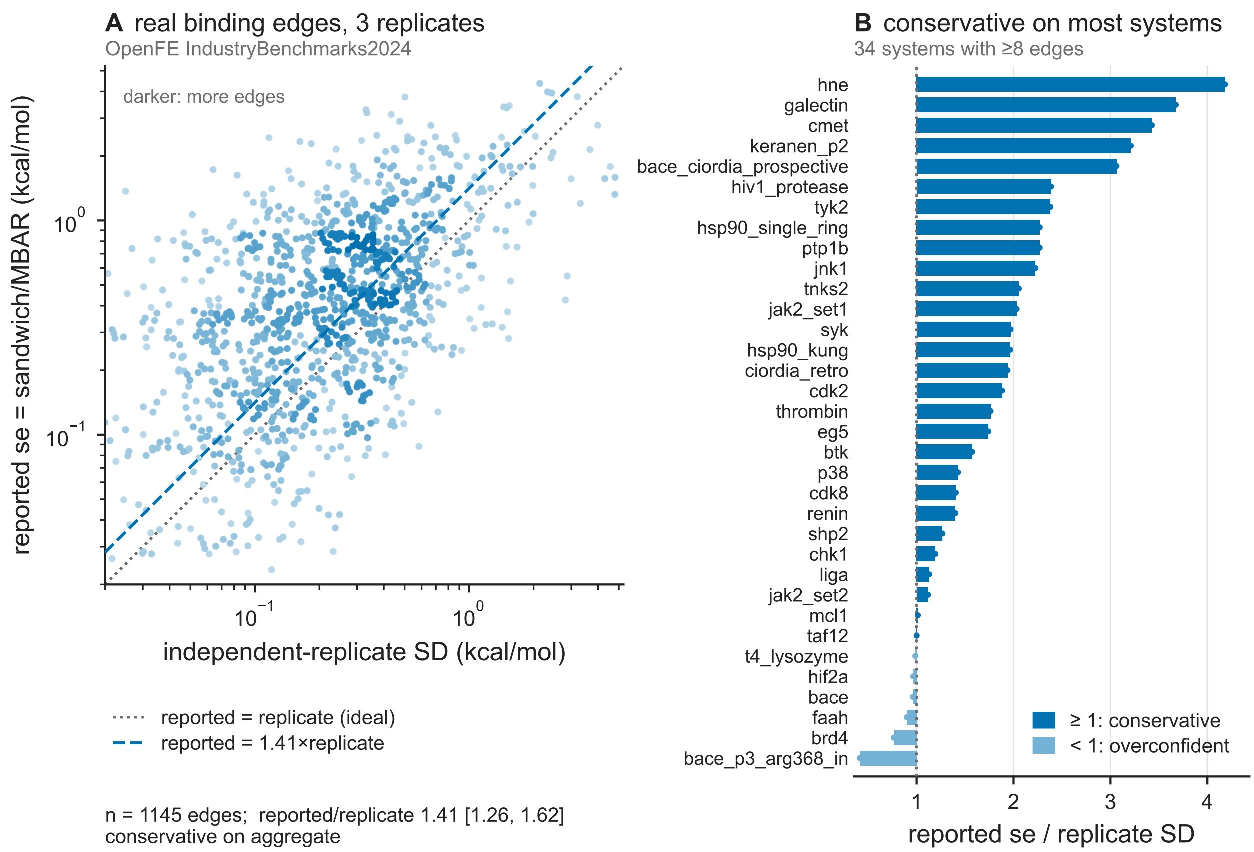

Figure 5: **Independent-replicate validation on real binding edges.** Reported per-edge standard error versus the across-replicate empirical standard deviation, on 1145 OpenFE Industry-Benchmarks2024 [2] edges (3 independent replicates each). (A) Reported standard error versus across-replicate standard deviation, with the $y{=}x$ line. (B) Per-system pooled ratio of reported standard error to across-replicate standard deviation, for the 34 benchmark systems carrying at least eight edges.

identify which of them is present, and it does not separate them from a fixed consequence of the one starting structure every repeat begins from.

The calibration underwriting the null is validated against seed and velocity replicate spread, a lower bound on full run-to-run reproducibility, since it excludes pose and system-preparation variance. That restriction bounds what the detection claims below can mean: an inconsistency that survives repeats sharing their starting coordinates is established as more than sampling noise, and not as physical error.

The test performs detection rather than weighting of a fixed estimate. The Gauss–Markov ceiling that makes the calibrated variance second-order for ranking, in a connected graph the GLS contrast mean being unbiased for any positive weights, does not bind a goodness-of-fit test, whose statistic depends on the weights directly.

A calibrated cycle-closure test. The fit of Section 3 gives residuals r_e and the goodness-of-fit statistic $X^2 = \sum_e r_e^2/V_e^{\text{rep}} \sim \chi^2(\text{dof})$, with dof the number of independent cycles (E minus the incidence rank). Under correctly-sampled physics $\mathbb{E}[X^2/\text{dof}] \approx 1$; here it runs below one, at a median reduced χ^2 of 0.34, qualitatively consistent with the conservative bars of Figure 5. A reduced χ^2 well above one signals edge-level inconsistency in excess of sampling noise, and the test is correctly blind to node-consistent force-field bias, which cancels around a loop.

Two variances have to be told apart from here on. The variance a practitioner is handed, V_e^{rep}, is what the fit uses as its weight; the variance the edge actually has is V_e^{true}. They are related by $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$, so the per-edge calibration ratio c_e is the reported standard error over the true one: $c_e > 1$ is a reported bar that is too wide (conservative) and $c_e < 1$ one that is too tight (overconfident). On this benchmark c_e is measured as the reported standard error over the across-replicate standard deviation.

Four summaries of that one distribution appear in this article. Each is a different functional of it, each is named here, and each is called by that name from here on. The *replicate ratio* is the pooled reported standard error over replicate spread, 1.41 (Figure 5). The *closure-weighted mean* is $\mathbb{E}[\chi_\nu^2]$, measured here at 0.85: whitening by the reported errors gives $\mathbb{E}[X^2] = \text{tr}(\Pi D)$ with $D = \text{diag}(c_e^{-2})$, so by Theorem 1 $\mathbb{E}[\chi_\nu^2]$ is the curl-leverage-weighted *mean* of c_e^{-2}. The *calibrated scale* is its square root, 0.92, the factor a reported bar has to be multiplied by to be right. The *inverted median* is one over the square root of the median reduced χ^2, and it recovers a scalar conservatism only when c_e is constant across edges, which Figure 5B shows it is not.

No single scalar conservatism factor is therefore identified by these data, and the apparent disagreement between the replicate ratio and the closure-weighted mean is an artefact of comparing different functionals of a heavy-tailed distribution, not a physical discrepancy. That distribution is drawn with the published summaries marked on it (Figure 6; `make figCal`): the per-edge ratio spans a factor of 128 between its first and ninety-ninth percentiles while the summaries occupy a 1.57× window. Part of that spread is guaranteed rather than measured, since a spread estimated from three replicates would put the same family of functionals between 0.71 and 2.88 even if every reported bar were exactly right.

Residual correlation among edges sharing a ligand endpoint, which the diagonal-V GLS does not model, is separately negligible (excess correlation over the projector-induced null -0.0079 $[-0.0314, +0.0164]$), and the Benjamini–Hochberg flag set is unchanged by it.

Network maximum-likelihood estimation of $\Delta\Delta G$ networks [31], cycle-closure hysteresis as a diagnostic [5, 29], weighted cycle closure [19], and the constrained variational solution of a whole network under loop closure [7] are established; what is new is the *calibrated* null, a per-edge aleatoric variance measured against seed and velocity repeats instead of assumed, which gives the test per-edge

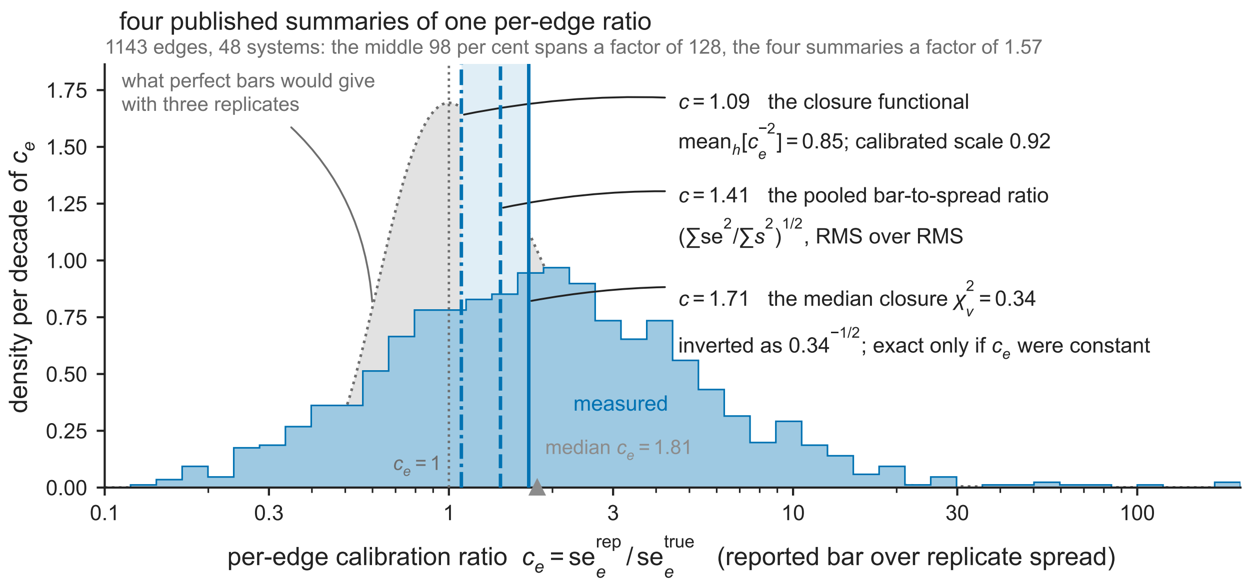

Figure 6: **One heavy-tailed per-edge ratio, and the summaries this article quotes from it.** The calibration ratio $c_e = \mathrm{se}_e^{\mathrm{rep}}/\mathrm{se}_e^{\mathrm{true}}$ over the 1143 edges of the 48 systems that carry a cycle, on a logarithmic axis because c_e is a ratio, so a bar twice too wide and one twice too tight are equally miscalibrated, and because the tail is the reason the summaries disagree rather than a nuisance to be clipped. Quartiles 0.91, 1.81 and 3.60. Each mark is a different functional of this one distribution: the curl-leverage-weighted mean of c_e^{-2} that Theorem 1 makes $\mathbb{E}[\chi^2_\nu]$, together with its root the calibrated scale, which are one mark and not two; the replicate ratio of Figure 5, the pooled reported standard error over replicate spread, 1.41; and the inverted median, the median reduced χ^2 of 0.34 inverted, which recovers a scalar conservatism only if c_e were constant across edges and carries that caveat on the figure. The grey silhouette is the distribution perfect bars would give at three replicates, where $s^2/\sigma^2 \sim \chi^2_2/2$ makes $c_e = 1/\sqrt{\mathrm{Exp}(1)}$ a parameter-free null with median 1.20: part of the measured spread is the denominator's own small-sample noise and only the excess is per-edge heterogeneity. The panel adds no measurement, and no claim beyond the text's: no single scalar conservatism is identified by these data. Reproduce with `make figCal`.

adaptive thresholds and complements the fixed hysteresis cutoffs in standard use. Those repeats share one pose and one system preparation, so the variance they validate excludes the component the alternative hypothesis is built from, no controlled false-positive rate follows from them, and nothing on this benchmark flags once the bars are widened to cover that component.

The calibrated null comes together with the cross-replicate reproduction check of Figure 3B and the estimation–detection conservation law of Theorem 1, which fixes which edges are auditable at all. What is shared with the works above is the weighted least-squares treatment of the cycle graph itself.

On the same 1143 edges a pre-registered head-to-head scored the calibrated test against the standard fixed 1.0 kcal/mol hysteresis cutoff and against a fixed-se χ^2 test, by area under the receiver-operating-characteristic curve; the criterion required beating both, and the verdict was a tie (Supplementary Figure S11). No further reading attaches to the ordering inside that tie, because the only label these data supply divides each residual by the same per-edge variance the calibrated rule uses as its weight: the ordering is monotone in proximity to that variance and not in detection skill, and the comparison ranks nothing.

The two rules are also not independent. On replicate 0 the three flag sets are strictly nested, the calibrated test flagging 6 systems, the fixed-se test 11 and the 1.0 kcal/mol cutoff 26, so the calibrated test catches nothing the cutoff misses and instead removes 20 of its flags.

The three-way chain does not survive the other repeats intact: on replicate 1 the calibrated test flags 6, the fixed-se test 13 and the cutoff 24, and two systems break it, `ciordia_retro` flagged by the calibrated test but not by the fixed-se test and `mup1` flagged by that test but not by the cutoff. On replicate 2 the chain holds again at 3, 11 and 24. The containment any claim here rests on is the two-set one, between the calibrated test and the cutoff, and it holds on all three replicates.

Whether removing those 20 flags is an improvement is what the contaminated label cannot say, and adjudicating them needs new sampling or new experiment, which public retrospective data cannot supply. The calibrated test's advantage over a fixed cutoff is accordingly structural and is not demonstrated here: per-edge adaptive thresholds, a null measured against seed and velocity repeats instead of assumed, and the observability map of Theorem 1. That measurement is not a controlled error rate, since the repeats hold pose and preparation fixed.

The null is conservative in the bulk, at the median reduced χ^2 above, so on a typical system the test under-flags. The median of the statistic does not bound the realized false-positive rate: Benjamini–Hochberg controls the false discovery rate through the behaviour of the null p-values near the rejection threshold and not through the median of the statistic, and the per-system calibration of Figure 5 is heterogeneous enough that a conservative bulk and an anti-conservative tail can coexist.

The correction is applied across systems, and the guarantee is carried by Benjamini–Yekutieli [3], which assumes no dependence structure at all, so the independence Benjamini–Hochberg needs is not required. It costs a factor $H_{48} = 4.46$ in the threshold and returns the identical six systems on replicate 0 and the identical three on replicate 2, dropping one, `brd4`, on replicate 1 (Supporting Information). Positive regression dependence is a plausible remark, and nothing here rests on it as an assumption: the systems are fitted separately and share no edges, though they share a force field and, within a target, a preparation.

The level quoted is nominal and not verified. The case for the flags therefore rests on the distribution-free cross-replicate reproduction check below rather than on the χ^2 tail; that check is independent of the χ^2 scaling and assumes no Gaussian null.

A subtler concern is that four of the six flagged systems are themselves locally overconfident against their own across-replicate spread (Figure 5: `brd4` $0.76\times$, `faah` $0.90\times$, `bace` and `hif2a` $0.96\times$), so a flag could partly reflect that system's own too-tight bars. Retesting each against the width its own replicates measure keeps all six at the per-system granularity, and two, `bace` and `cdk8`,

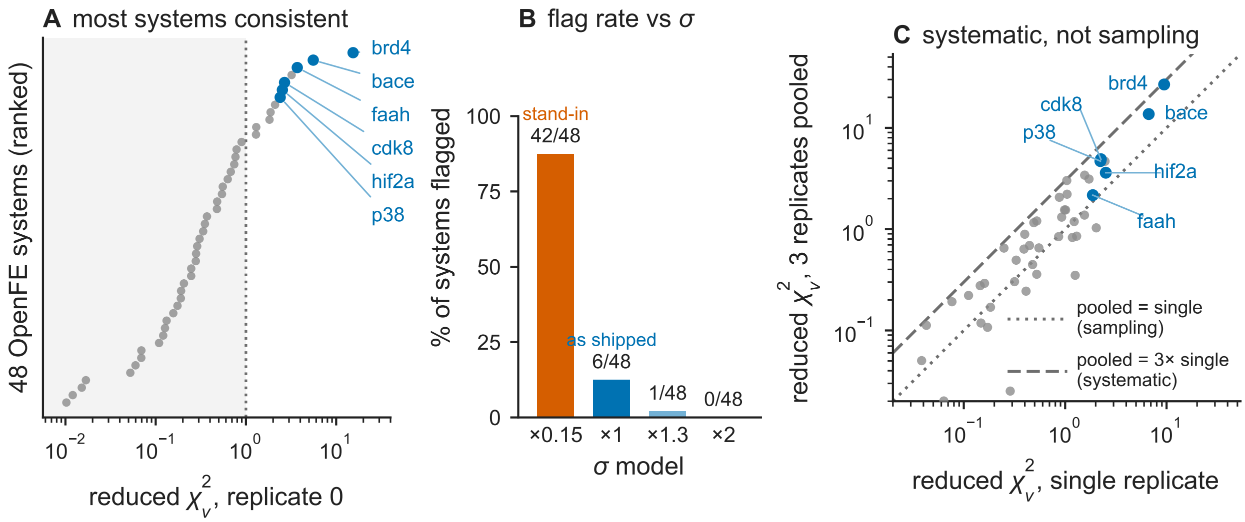

Figure 7: **Calibrated cycle-closure quality control.** (A) Per-system reduced χ^2 from a generalized-least-squares cycle-closure fit, ranked, with shading below $\chi^2_\nu = 1$ and the flagged systems marked. (B) Percentage of systems flagged under four σ models, in plotted order: a uniform $\times 0.15$ shrink, the reported bar as shipped, a $\times 1.3$ inflation and a $\times 2$ inflation. The shipped bar is the tick marked $\times 1$; the scale at which those bars are correct is measured as 0.92 of them, which is a different quantity from the tick. (C) Reduced χ^2 from a single replicate against the value from the three replicates pooled, with reference lines at pooled/single $= 1$ and $= 3$.

at the per-edge granularity this work pre-specified. The per-edge arm is the only self-calibrated arm holding its level (1/400, that is 0.003 with an exact binomial interval $[0.000, 0.014]$, against a nominal 0.05), and so the conservative answer (Supplementary Table S8).

The test on the replicate benchmark. On the OpenFE IndustryBenchmarks2024 [2] (1145 edges over 49 systems, of which the 48 with ≥ 1 independent cycle, holding 1143 edges, admit the test; 3 replicates), the median reduced χ^2 is 0.34 and a Benjamini–Hochberg false-discovery-rate (FDR) test at 0.05 applied to the first replicate flags six chemically-sensible systems (`brd4`, `bace`, `faah`, `cdk8`, `hif2a`, `p38`; Figure 7A; `brd4`'s data set is a buried-water set, `bace` is protonation-sensitive). That particular set of six is a single-replicate draw and should not be read as a census; Section 6.1 measures how much of it survives the other two replicates.

One of the six has an identifiable cause. The benchmark labels a system by target, and eight of its names union edges from more than one separately released data set. Where those releases share ligands the union carries cycles present in neither release alone, and the cycle spaces sit orthogonally inside the union's, so the closure statistic splits exactly. On `cdk8` the split is decisive: the union gives $\chi^2 = 77.7$ on 29 degrees of freedom, of which the two releases carry 31.1 on 23 between them, leaving 46.6 on the six cycles that span them, a reduced χ^2 of 7.8 at $p = 2.3 \times 10^{-8}$. Neither release is inconsistent on its own: `merck_cdk8` (51 edges, 20 degrees of freedom) gives $p = 0.218$, 0.865 and 0.189 on the three replicates, and `misc_cdk8` (12 edges, 3 degrees of freedom) gives 0.091, 0.992 and 0.985, the first being the smallest of the six and still well short of any threshold used here. The spanning cycles exist because the two releases share seven ligand *names*, and all seven denote chemically different molecules in the two releases, name `13` being a benzyl-indazole carboxamide in one and a chlorophenoxy-thienopyridine carboxamide in the other. Every spanning cycle therefore asserts that two distinct ligands have the same free energy, and resolving each edge within its own release removes all six on every replicate, taking the degrees of freedom from 29 to the within-release total of 23. The 46.6 is a bookkeeping collision in the benchmark's ligand naming and not a disagreement between two preparations. `cdk8` is flagged for that collision, and the flag is retained in the published set with its cause named, because a system whose network asserts a false identity is what an inconsistency test should reject.

The hazard belongs to the analysis and not to the released data. Each release is internally consistent and each names its own ligands unambiguously; what fails is the join, and it fails for anyone who unions two releases of a target on ligand names without checking that the names denote the same molecules. That is a common enough way to assemble a benchmark network that it is worth saying plainly: match on structure, not on name.

A closure statistic reads the network it is given, and an inconsistency produced by the bookkeeping that assembled that network enters as the same cyclic residual as one produced by the simulations. No other system carries across-release cycles that matter. Two shared ligands are needed before a union carries a cycle that spans its releases, and in six of the eight the releases share at most one, so the union is disconnected or joined by a tree and no such cycle exists at any magnitude; `tyk2`, whose two shared names do denote the same molecules, gives one spanning cycle that is unremarkable ($p = 0.67$). The single cross-release comparison that joins matching ligands shows no excess.

What the benchmark supplies on between-preparation variance is therefore a negative result rather than a number. With `cdk8`'s six spanning cycles resolved away as a naming collision, the only genuine cross-release join anywhere in the benchmark is `tyk2`'s single cycle, which never departs from its null ($p = 0.67$, 0.40 and 0.84 on the three replicates) and whose across-release block never rises above 0.64 of the within-release one (Supplementary Figure S8; `make figRelease`). The public data contain no usable estimate of what a second preparation of the same target adds, which bounds what can be said about the pose and preparation component throughout this article rather than supplying it. Controlled work does bound it from outside. Starting one alchemical edge from two kinetically distinct conformations of the same protein shifts its free energy by as much as 4.74 kcal/mol under a default protocol, a root-mean-square inconsistency of 4 that falls to 0.57 only once the runs are extended to 55 ns, and the effect is put at 2–5 kcal/mol in general [20]; independently, careful preparation of protein and ligand structures is what brings a leading workflow to the reproducibility of the experiments it is scored against [24]. The mechanism is kinetic trapping rather than thermal averaging, so a wrong starting state persists as a fixed offset for the whole run, which is why repeats that share it cannot price it.

The grouping is therefore a choice the analysis makes, and the flag set is not invariant to it. Re-running the same replicate-0 test on the release's own blocks gives 56 testable systems and again six flags, but a different six: `cdk8` leaves and `jacs_mcl1` enters. The grouping by target name was not pre-registered, and the Benjamini–Hochberg family size depends on it. The published list is the by-target one throughout. Supplementary Tables S6 and S7 give the per-replicate degrees of freedom, reduced χ^2 and adjusted q-value behind every flag.

The test's selectivity is entirely a function of calibration (Figure 7B). A uniform $\times 0.15$ shrink standing in for the learned head's central overconfidence (Supplementary Figure S2) flags 88% of systems (42 of 48), which leaves almost nothing unflagged and so no longer separates anything; the reported bar as shipped flags 12% (6 of 48), a $\times 1.3$ inflation one system of 48, and a $\times 2$ inflation none. These are flagged fractions and not false-positive rates: which of the flags are false is not known here (Section 8).

The count is not separable from the level, and the shipped operating point is the conservative one. The scale at which the reported bars are correct is the calibrated scale, which follows from the closure-weighted mean, the functional that governs the statistic, and not from a pooled ratio; at that mean the bars are wide by 1.09×. Section 6.2 puts the closure-weighted mean in $[0.63, 1.08]$, and the calibrated scale accordingly in $[0.79, 1.04]$. Correcting every reported bar to a scale in that interval and re-running the test on the real data gives six flags at 1.04 and at 1.00, seven at 0.92 and nine at 0.79 (`make figSelf`).

Two scales have to be kept apart here, and they are reciprocal. The calibrated scale is $\mathrm{se}^{\mathrm{true}}/\mathrm{se}^{\mathrm{rep}}$, the reciprocal of the per-edge ratio c_e, measured above as 0.92; a simulation that keeps the test on the reported bars must therefore draw its truth at that many times them, which is a null 1/0.92 times the truth. Quoted on the calibrated scale, the realized probability that the test flags anything when nothing is wrong is 0/400 at 0.79, 1/400 at 0.92, 18/400 at 1.00 and 40/400 at 1.04. As levels with exact binomial intervals those are 0.000 $[0, 0.009]$, 0.003 $[0.000, 0.014]$, 0.045 $[0.027, 0.070]$ and 0.100 $[0.072, 0.134]$. Four hundred draws do not separate the two lower points from each other or from zero, so nothing is read from their ordering. What the grid supports is the upper end, where the interval clears the nominal level, and the direction between the ends.

That simulation verifies the Benjamini–Hochberg implementation and not the real-data operating point, which is why the level quoted above is nominal and not verified. What it does establish is a direction: the level sits below nominal at the measured calibration and above it at the other end. Because the bars are measured wide, the level across the calibration interval runs from essentially zero at 0.79 to 0.100 at 1.04, so the exposure that matters is the anti-conservative end, where a bar slightly tighter than the truth doubles the nominal rate.

Sensitivity is exposed in the opposite direction. At 1.3× one system of 48 flags and at 1.5× and 2× none do, so nothing on this benchmark survives a bar widened to cover pose and preparation variance. Widening changes the null being tested instead of moving the same test along it, and what is fragile on that side is sensitivity and not validity.

The uniform shrink is a stress test and not a realistic head, and the comparison is a counterfactual in a further sense: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check. The head's own measured, overlap-dependent profile applied per edge flags 42 of 48, so the degradation is not an artifact of uniformity, and a shuffled control with a swept magnitude shows it is the magnitude that destroys the test while over-wide bars cost power the other way (Supporting Information). The demonstration is accordingly scoped to heads as overconfident as the measured one (reported se 5–11× too small): it establishes that such heads saturate the test, and not the shrink magnitude at which selectivity is first lost.

The three independent replicates confirm the flagged inconsistency is reproducible and not sampling. Pooling the replicates shrinks the bars (se $\to$ se$/\sqrt{3}$) yet increases the flagged systems' reduced χ^2 toward the 3× variance-components ceiling (Figure 7C), which occurs only for a component that repeats (single-replicate $\chi^2_\nu = 1 + \sigma^2_{\mathrm{sys}}/V_{\mathrm{samp}}$ vs pooled $1 + 3\sigma^2_{\mathrm{sys}}/V_{\mathrm{samp}}$, with σ^2_{sys} the reproducible variance component and V_{samp} the sampling one; the ratio $\to 3$ iff reproducible-dominated). The argument fixes the component's variance and not its origin: it is invariant to whether σ^2_{sys} is physical error or a fixed consequence of the one starting structure the three repeats share, both being constant across repeats.

Cross-replicate reproduction of the flagged inconsistency. Greedily removing the flagged (highest-$|z|$) edges restores consistency (reduced $\chi^2 \leq 1$) in 3–7 edges, versus 12–37 removed at random (Figure 3A). That contrast illustrates the repair procedure and is not evidence, because it cannot fail: the statistic being driven down is $X^2 = \sum_e r_e^2/V_e^{\mathrm{rep}}$ and the removal order is descending z_e^2, that sum's own per-edge summand, so a network with no localized systematic error at all reproduces the same contrast.

The evidential check is instead out-of-sample. Fitting each replicate's network separately, the per-edge standardized residuals correlate across independent replicates ($r = +0.30$ to $+0.42$, $n = 1143$, each interval excluding zero under a bootstrap over systems; Figure 3B), where sampling noise alone would give zero. What the flags identify is therefore reproducible across runs and not a property of one draw. The correlation carries the same restriction as the pooling argument: any per-edge quantity held fixed across the three repeats produces it, the shared starting structure is such a quantity, and what is established is reproducibility and not a physical origin.

Summary. The calibrated bar supplies, at no label cost, the null that turns cycle closure into a test separating reproducible from sampling inconsistency, and a bar as overconfident as the learned σ measured here destroys it. Theorem 2 bounds what that test, or any other of its kind, can be asked for, which is why its flags are not a repair list.

6.1 What survives a second replicate

The benchmark ships three independent repeats, so the flag set can be recomputed instead of assumed, and recomputing it shows that the set does not reproduce.

Two conventions bound how the union below should be read. It is descriptive and carries no controlled error rate. Three tests each holding a nominal 0.05 would put the chance of at least one false flag among them at $1 - 0.95^3 = 0.143$ if they were independent, and these three are not; Benjamini–Hochberg in any case controls the false discovery rate rather than the probability of any false flag. The counts are also per system and not per target, the benchmark's 49 systems carrying 41 protein labels, with `bace` contributing four and `hsp90` four; the six flagged systems are six different proteins, so that count at least is not inflated by system splitting.

The same pipeline at the same FDR level returns {`bace`, `brd4`, `cdk8`, `faah`, `hif2a`, `p38`} on the first replicate, {`bace`, `brd4`, `ciordia_retro`, `hif2a`, `p38`, `renin`} on the second and {`bace`, `cdk8`, `renin`} on the third: eight systems are flagged at least once, one (`bace`) is flagged every time, and pairwise Jaccard overlap is 0.50, 0.29, 0.29 against 0.056 for size-matched random draws, averaged over the three pairs. That instability is not a threshold artefact and persists from $\alpha = 0.01$ to $\alpha = 0.20$ (Supplementary Figure S12). Pooling the three replicates flags seven systems, the union minus `faah`.

The instability is not explained by how cyclic the networks are. Cycle counts on this benchmark are low: nine of the 48 systems carry exactly one independent cycle, twenty-three carry three or fewer, and the quartiles are 2, 5 and 12. A reduced χ^2 on one or two degrees of freedom is an unstable statistic, and those systems do vary more across repeats. But they are not the systems that move in and out of the flag set: seven of the eight ever flagged are well connected, and restricting the comparison to the best-connected third of the benchmark leaves the pairwise overlap no better than it is over all 48. The instability sits where the test has power, which makes it a property of the measurement rather than of thin networks.

Two properties do reproduce. The ranking of systems by closure inconsistency reproduces across replicates (Spearman 0.649, 0.644, 0.723 over all 48 systems), and the dependence of selectivity on σ-calibration, which is the claim this section makes, is essentially unmoved: the $\times 0.15$ shrink flags 42, 41, 41 systems, the reported bar 6, 6, 3, a $\times 1.3$ inflation 1, 1, 1, and a $\times 2$ inflation none, on the three replicates respectively. A single replicate therefore supports a ranking and a calibrated operating point, and not a list of names.

Among the published six, `faah` is the weakest on every measure. Its reduced χ^2 crosses one in both directions across replicates (3.71, 0.45, 1.47), its standardized residuals barely reproduce between replicates ($r = -0.085$), and it is the one member the pooled fit drops: a flag a second replicate does not support, and not a member of the set on the footing of `bace`.

6.2 Out-of-sample behaviour

Two checks use one replicate to predict another and so cannot be circular. First, the per-edge miscalibration measured from replicates two and three predicts the closure statistic of the replicate they exclude. With $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$ as above, Theorem 1 makes the closure-weighted mean predictable from c_e alone, and that prediction correlates with the observed per-system χ^2_ν at Spearman 0.648 ($n = 48$). Both sides divide by the same reported variance, so the reference is not zero: under a simulation with no systematic error anywhere the same statistic returns 0.059 with a 95% upper limit of 0.364, and 0.648 exceeds every one of 200 draws (Supplementary Figure S13A).

The aggregate level agrees less well. The predicted closure-weighted mean is 0.85, with interval $[0.63, 1.08]$. The observed degree-of-freedom-weighted value is 1.32, with interval $[0.81, 1.80]$. The two intervals overlap, and the prediction does not cover the observation.

The scope of that agreement is narrow. The prediction is built entirely from measured miscalibration, so a rank agreement of this size says that the closure statistic largely reads out how well each system's bars are scaled. It is a statement about what the test responds to, and not evidence that the flagged systems are in error. The flagged systems do sit above the line, which is what any reproducible offset does: it inflates a closure residual without inflating the spread between repeats. Being reproducible across repeats that share a starting structure is what places a system above the line, and that is weaker than being in error.

Second, edges selected on one replicate are more inconsistent on the two held out than equal-size random selections are (mean gain in held-out reduced χ^2, Stouffer $p = 1.3 \times 10^{-3}$, 1.3×10^{-3}, 4.4×10^{-3} over the three rotations), while a topology-only control built from curl-leverage is indistinguishable from random ($p = 0.21$–0.91). Each candidate statistic was calibrated against a no-error null before use and the realized level of every one is reported; two fired far above nominal and are not used. Only the consistency-gain statistic is specific to the flagged systems: an $|z|$-rank-matched control drawn from unflagged systems reproduces the leverage-normalised residual signal but not the consistency gain. The effect is an aggregate one, significant individually in two of the five testable systems.

7 Computational details

Data. Every closure fit consumes the release's combined per-replicate table directly, taking the per-leg complex and solvent ΔG and $\mathrm{d}\Delta G$ columns and composing the edge value and its standard error from them; no network-level or cycle-closure correction is applied to an edge before it enters the fit. Figure 4 uses the `alchemtest` BACE1 relative-binding and benzene sets, from which forward and reverse work samples are available per λ window. The replicate validation, the cycle-closure test and every experiment on real binding data use the OpenFE IndustryBenchmarks2024 release [2], which ships three independent protocol repeats per transformation: 1145 edges across 49 benchmark systems. A system here is one of the benchmark's named edge networks and not one protein, `jak2_set1` and `jak2_set2` being separate systems built on the same protein, so the system count exceeds the number of distinct proteins (Section 6.1). The 48 systems with at least one independent cycle, which between them hold 1143 of the 1145 edges, admit the closure test, and each counts as one hypothesis in the Benjamini–Hochberg correction; the single remaining system is acyclic and contributes the other two edges. The per-system calibration ratios of Figure 5B are pooled instead over the 34 systems that carry at least eight edges, eight being the smallest edge count at which a ratio pooled against an $n = 3$ replicate spread is stable enough to plot on its own row; the other fifteen systems enter the aggregate ratio and the per-edge panel but are not shown individually. Grounding against measured affinity uses public ChEMBL IC_{50} values. The downstream uses audited in Section 8 run instead on the public OpenFF protein–ligand benchmark [9], which aggregates the FEP+ and Merck data sets [25, 29] and carries curated per-edge experimental $\Delta\Delta G$ labels, under both-endpoint scaffold-disjoint out-of-distribution splits; interval sharpness, commit-to-synthesis selection, reward re-ranking and the commit-trust decision run on it, while edge ranking and the stopping decision run in controlled simulations. No free-energy simulation was run for this work; all alchemical output is taken as published.

Per-edge variance. $\widehat{\Delta F}$ is obtained by solving Bennett's estimating equation $S(\Delta F) = 0$ by scalar root-find, and its variance from the closed form B/I^2 of Theorem 3 evaluated at the root. The variance is therefore computed from the same simulation output as the estimate, at no label cost.

Cycle-closure test. Node potentials are fitted by generalized least squares on the signed incidence matrix of the perturbation network, whitened by the per-edge standard errors, so that the fit is a weighted least-squares problem with conductances $1/V_e^{\mathrm{rep}}$. Writing N for that incidence matrix, to keep B for the sandwich meat of Theorem 3, the statistic is the reduced χ^2 of the residuals with $\mathrm{dof} = E - \mathrm{rank}(N)$; per-system tail probabilities use the exact regularized upper incomplete gamma function rather than a Wilson–Hilferty approximation, and systems are flagged by Benjamini–Hochberg control of the false discovery rate at $\alpha = 0.05$ across systems. The population for the closure test is the 48 systems that carry at least three replicate-0 edges and at least one independent cycle, which between them hold 1143 of the benchmark's 1145 edges.

The two baseline detectors. The fixed hysteresis cutoff flags a system when the largest absolute closure sum over a fundamental cycle basis exceeds 1.0 kcal/mol. That basis is the cotree of a spanning tree of the perturbation network, so it holds one cycle per degree of freedom of the closure fit, and the aggregation over them is a maximum. The rule consumes no per-edge uncertainty and applies no multiplicity correction. The fixed-se test is the closure fit above with one standard error per system, the median of that system's per-edge reported values, substituted for every edge; its per-system tail probabilities pass through the same Benjamini–Hochberg control at $\alpha = 0.05$. The three detectors accordingly differ only in what they do with the per-edge bar: the cutoff ignores it, the fixed-se test replaces it by one number per system, and the calibrated test uses all of them.

Simulation protocols and model architectures. Two protocols underwrite numbers quoted above: the pure-null simulation that supplies every realized false-flag level, and the reproducible-systematic magnitude used as the label in detection comparisons. The amortized predictor used in the acquisition, stopping and reward experiments, whose predictive uncertainty $\sigma_{\mathrm{total}} = \gamma\sqrt{\sigma_{\mathrm{ens}}^2 + \sigma_{\mathrm{ale}}^2}$ composes deep-ensemble epistemic spread with a benchmark aleatoric floor and is conformally scaled at miscoverage $\alpha = 0.1$, and the learned-variance baseline of Supplementary Figure S2 are separate and much smaller models. Both protocols, and the features, layer sizes, objectives and training budgets of both models, are specified in the Supporting Information.

Software. Python $\geq$ 3.11 with numpy 2.5.0, scipy 1.18.0, pymbar 4.0.3, torch 2.12.1, RDKit 2026.3.3 and matplotlib 3.11.0. The pipeline is deterministic: in the environment recorded here, re-running a figure's command reproduces that figure's rendered raster byte for byte, and this was checked on the figures that depend on ensemble training rather than assumed. Ensemble training runs full batch on CPU for a fixed number of epochs, each member is seeded from its own index, and the ligand ordering reaching the featurizer and the sampler is sorted rather than read out of a set, so no unrecorded source of variation enters the training path.

8 Limitations

The sandwich is aleatoric-only and conditional on correctly sampled ensembles. Systematic force-field bias lies outside it, and the out-of-distribution behaviour audited here rests on an ensemble plus a scalar aleatoric floor: there is no learned force-field-correction head, so the three-way decomposition of the synthetic illustration is not reproduced on real data.

The binding-edge se is validated against three independent replicates (Figure 5). In aggregate the replicate ratio is $\sim$1.41$\times$ and the bar is conservative, but calibration is heterogeneous: of the 34 systems carrying at least eight edges, 28 are conservative and six overconfident, namely `t4_lysozyme` 0.99$\times$, `hif2a` and `bace` 0.96$\times$, `faah` 0.90$\times$, `brd4` 0.76$\times$, and the protonation variant `bace_p3_arg368_in` 0.41$\times$, the only one of the six that is not marginal. Those replicates are seed and velocity repeats from identical starting coordinates, so the across-replicate SD is a lower bound on full reproducibility, excluding pose and preparation variance, and the measured conservatism may be closer to neutral against real reproducibility. The same restriction limits what the quality-control test detects: a pose or a protonation assignment carried unchanged into all three repeats reproduces exactly as physical error does, so the flags are reported as reproducible inconsistency throughout.

The test detects only *edge-level* inconsistency, and neither removal rule improves accuracy against experiment: guided removal ties random on the four grounded systems (Stouffer $p = 0.48$; Figure S10) and the influence rule does no better overall, doing measurable harm on `p38` and reaching a nominal improvement on `bace`, neither clearing the conjunctive criterion (Section 4; Figure 2C). Theorem 2 offers a structural reading of both nulls, bounded by the margin Section 4 measures and not established independently of it. The pre-registered falsifier for that reading is a guided removal beating its random null in a majority of grounded systems at a combined p below 0.05, which would show the visible part actionable; it is not met.

The measurement rests on four systems selected by the test, so it characterises those systems rather than the benchmark, and for `cdk8` it characterises less even than that. Its grounded sub-network is all 31 edges of `merck_cdk8`, the block whose own closure statistic is not significant on any replicate, so the evidence that flagged it lies outside the sub-network measured. `p38`'s grounded sub-network is the `jacs_p38` block, which is individually significant ($\chi^2_\nu = 2.63$, $p = 1.9 \times 10^{-4}$), and `hif2a` and `bace` are single-block systems.

The measurement also runs on a connectivity-based ChEMBL join that is stereo-blind, so enantiomer pairs sharing a 2D graph collapse to one experimental value. That inflates absolute MUE while cancelling in the guided-versus-random contrast, since both arms read the identical cached affinity. The experimental and annotation errors the join carries are per-ligand quantities entering the same unauditable subspace as force-field bias, so the measurement locates the error without attributing it.

The influence rule is a second pre-registered hypothesis on the data that returned the first null, so it is a second look and not independent confirmation. The theorem bounds what a network's own edge values can support and says nothing against methods bringing information from outside the network. A prospective flag, repair and remeasure test with new molecular dynamics lies outside what public data can support.

The cross-replicate check shows that what is flagged reproduces across independent runs; it is not a false-positive-rate guarantee and does not certify the nominal Benjamini–Hochberg rate. Recall cannot be measured at all, there being no ground-truth error labels. Replacing each system's reported standard error by its own replicate-measured calibration changes the flag set in both directions, by an amount depending on the granularity and the direction, from retaining two of the six to reaching sixteen. The arms, their realized false-flag probabilities and the single one that holds its level are tabulated in the Supporting Information, and none is reported here as a flag set. The flag set is therefore sensitive to calibration error of the size actually present, in both directions, which limits how far any single flag list should be trusted.

Together with the median reduced χ^2, which implies limited single-replicate power, that makes the flagged systems a floor rather than a prevalence estimate. `renin` carries the reproducible signature by the pooled-replicate ratio, its reduced χ^2 rising to 4.7 pooled from a per-replicate mean of 2.5, and clears the false-discovery threshold on two of the three replicates though not on the first, which is the one the published list is drawn from; it is a system that list misses rather than one the test misses, and `ciordia_retro` is a second such case.

A calibrated uncertainty was audited on six downstream uses: interval sharpness, edge ranking in active learning, commit-to-synthesis selection, reward re-ranking over a measured congeneric series, and two decision tasks, commit trust and calibrated stopping. The quantity under test there is not the per-edge sandwich bar of Section 5 but the predictive uncertainty of an amortized $\Delta\Delta G$ predictor, a conformally scaled combination of ensemble spread and an aleatoric floor; the Supporting Information defines it. It leads a fair baseline in none of the six, and on one it does harm: the calibrated reward lowers the true hit-rate of a generator on the single evaluable congeneric series, because the uncertainty there tracks chemical novelty rather than quality. On commit trust the physics σ matches post-hoc recalibration by temperature scaling [8] or isotonic regression [32] rather than beating it, and the separation in the stopping simulation is absent at the six per cent miscalibration a same-budget pooled estimator reaches. Calibration is a precondition for decisions of this kind rather than a multiplier on their quality. Supplementary Table S11 collects every effect against its baseline.

The stopping experiment is a controlled simulation, and the reward re-ranking audit rests on a single σ-varying target, so its conclusion that σ tracks novelty rather than quality is directional and should be tested across more chemotypes.

9 Conclusion

A perturbation network's own edge values determine only the cycle part of its error. Fitting cycle closure against the precisions the free-energy estimator already reports turns a descriptive consistency check into a test with a stated null, orders networks by how far their own calculations disagree, and names in advance the edges that lie in no cycle and can carry no evidence. The test consumes only quantities the estimator computes anyway, and its selectivity is set by how well those quantities are calibrated. On these data the test catches nothing the fixed hysteresis cutoff misses: the two flag sets are nested and the head-to-head between them is a tie, so the advantage is structural, being per-edge adaptive thresholds, a null measured against seed and velocity repeats, and the observability map. The flags are also not a repair list: removing the edges they name does not recover accuracy against experiment, and a rule built from the theorem does no better and on one system does harm, because the component that sets accuracy against experiment lies in the gradient part, which no closure statistic reaches.

Those limits compound in the flag set this article publishes, and the sum is worth stating once.

Of the six systems flagged on the first replicate, one carries its flag from a naming collision created by the grouping the analysis chose, one is the weakest on every measure applied to it and drops when the replicates are pooled, and the membership of the set does not reproduce across the three repeats. What survives all three is the ordering of systems and the per-edge residuals behind it, which is what a reader should take the test to deliver.

Closing that gap needs information the network does not contain, and in absolute measurements the worst case costs one ligand per dimension. That is a worst case: a bias smooth in chemical space is a far smaller object, and a noisy anchor resolves only to its own precision.

A second ceiling comes from the measurement rather than from the decomposition. The bar the test uses as its null is measured against repeats that resample seeds and velocities while holding one pose and one system preparation fixed, so it prices sampling noise and not the preparation an entire network inherits. Widening it by half already leaves nothing on this benchmark flagged, and half is on the evidence an under-estimate of what that component costs: starting one alchemical edge from two kinetically distinct protein conformations moves its free energy by up to 4.74 kcal/mol, and the same study puts the general effect at 2–5 [20]. The test is therefore bounded twice: by the geometry of the network, and by a variance component its own null never sees. The practical use of cycle closure is accordingly as a calibrated alarm, with the error that moves a ranking to be sought in absolute measurements or independent preparations instead.

Acknowledgements

The author thanks colleagues at the Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University, and at the Institute of Biomedical Chemistry for discussions. This research received no external funding.

This work was prepared with the use of Anthropic's Claude, run through the Claude Code interface, for the code, the analyses, the figures and the drafting; the author verified the results and takes responsibility for them.

Author Contributions

N.L.P. is the sole author and is responsible for the conception and design of the study, the theory, the software, the analysis, the preparation of the figures, and the writing of the manuscript.

Conflicts of Interest

The author declares no competing financial interest.

Data and Software Availability

All data used in this work are public. The alchemtest BACE1 and benzene alchemical sets supply the work samples of Figure 4; the OpenFE IndustryBenchmarks2024 replicate set supplies the 1145 binding edges and their three protocol repeats; the OpenFF protein–ligand benchmark, which aggregates the FEP+ and Merck data sets, supplies the downstream audit tasks; and ChEMBL supplies the experimental affinities used for grounding. No free-energy simulation was run for this work; all alchemical output is taken as published. Two findings reported here were sent to the maintainers of that benchmark as public issues on its repository, `https://github.com/OpenFreeEnergy/IndustryBenchmarks2024`: the per-edge auditability and detectable-shift tables of Section 3 as issue 331 on 9 August 2026, and the `cdk8` ligand-name collision of Section 6 as issue 332 on 22 August 2026. On the second the maintainers confirmed the collision, reported that their own analyses were unaffected by it, and added a note to the release's analysis documentation recording that ligand names are not unique across data sets; the first remains open. The BAR layer, the cycle-closure quality-control module, the theorem-verification and quality-control unit tests, and every figure-regeneration script are released as open source under a permissive licence, with each figure in this article regenerated by a single documented command. The code is available at `https://github.com/doctawho42/fep-surrogate-pipeline` under the MIT licence; the version from which every figure in this article was generated is tagged `v1.3.0`. The archive is deposited at `https://doi.org/10.5281/zenodo.22046692`, a concept identifier that resolves to the most recently deposited version.

References

[1] Anastasios N. Angelopoulos and Stephen Bates. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv preprint, 2021.

[2] Hannah M. Baumann, Joshua T. Horton, Michael M. Henry, Alyssa Travitz, Benjamin Ries, Richard J. Gowers, David W. H. Swenson, Iván Pulido, Dominic Rufa, David L. Dotson, Nupur Bansal, Joseph P. Bluck, Howard Broughton, Kira Campbell, Lili Cao, Benedikt Frieg, Vytautas Gapsys, Hendrik Göddeke, Marco Klähn, Sirish Kaushik Lakkaraju, Stephanie M. Linker, Thomas Löhr, Aniket Magarkar, Sergio Pérez-Conesa, Hans E. Purkey, Hayk Saribekyan, Jenke Scheen, Christina E. M. Schindler, Thomas Steinbrecher, Chaya D. Stern, Patricia Suriana, William C. Swope, Gary Tresadern, Lev Tsidilkovski, Binqing Wei, Alexander H. Williams, Yao Wu, Ivy Zhang, John D. Chodera, James R. B. Eastwood, David L. Mobley, and Irfan Alibay. Large-Scale Collaborative Assessment of Binding Free Energy Calculations for Drug Discovery Using OpenFE. *Journal of Chemical Information and Modeling*, 66(11):6429–6452, 2026. doi: 10.1021/acs.jcim.6c00089.

[3] Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 2001. doi: 10.1214/aos/1013699998.

[4] Charles H Bennett. Efficient estimation of free energy differences from Monte Carlo data. *Journal of Computational Physics*, 22:245–268, 1976. doi: 10.1016/0021-9991(76)90078-4.

[5] Di Cui, Bin W. Zhang, Zhiqiang Tan, and Ronald M. Levy. Ligand Binding Thermodynamic Cycles: Hysteresis, the Locally Weighted Histogram Analysis Method, and the Overlapping States Matrix. *Journal of Chemical Theory and Computation*, 16(1):67–79, 2020. doi: 10.1021/acs.jctc.9b00740.

[6] Sofia Dias, Nicky J. Welton, Valéria C. C. Marinho, Georgia Salanti, Julian P. T. Higgins, and A. E. Ades. Estimation and Adjustment of Bias in Randomized Evidence by Using Mixed Treatment Comparison Meta-Analysis. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 173(3):613–629, 2010. doi: 10.1111/j.1467-985x.2010.00639.x.

[7] Timothy J. Giese and Darrin M. York. Variational Method for Networkwide Analysis of Relative Ligand Binding Free Energies with Loop Closure and Experimental Constraints. *Journal of Chemical Theory and Computation*, 17(3):1326–1336, 2021. doi: 10.1021/acs.jctc.0c01219.

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *International Conference on Machine Learning (ICML)*, 2017.

[9] David Hahn, Christopher Bayly, Melissa L. Boby, Hannah Bruce Macdonald, John Chodera, Vytautas Gapsys, Antonia Mey, David Mobley, Laura Perez Benito, Christina Schindler, Gary Tresadern, and Gregory Warren. Best Practices for Constructing, Preparing, and Evaluating Protein-Ligand Binding Affinity Benchmarks [Article v1.0]. *Living Journal of Computational Molecular Science*, 4:1497, 2022. doi: 10.33011/livecoms.4.1.1497.

[10] J. P. T. Higgins, D. Jackson, J. K. Barrett, G. Lu, A. E. Ades, and I. R. White. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Research Synthesis Methods*, 3(2):98–110, 2012. doi: 10.1002/jrsm.1044.

[11] David C. Hoaglin and Roy E. Welsch. The Hat Matrix in Regression and ANOVA. *The American Statistician*, 32(1):17–22, 1978. doi: 10.1080/00031305.1978.10479237.

[12] Peter J. Huber. The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1*, pages 221–233. University of California Press, 1967.

[13] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye. Statistical ranking and combinatorial Hodge theory. *Mathematical Programming*, 127(1):203–244, 2011. doi: 10.1007/s10107-010-0419-x.

[14] Tuomo Kalliokoski, Christian Kramer, Anna Vulpetti, and Peter Gedeck. Comparability of mixed IC_{50} data – a statistical analysis. *PLoS ONE*, 8(4):e61007, 2013. doi: 10.1371/journal.pone.0061007.

[15] Alex Kendall and Yarin Gal. What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[16] D. J. Klein and M. Randić. Resistance distance. *Journal of Mathematical Chemistry*, 12:81–95, 1993. doi: 10.1007/BF01164627.

[17] Ulrike Krahn, Harald Binder, and Jochem König. A graphical tool for locating inconsistency in network meta-analyses. *BMC Medical Research Methodology*, 13:35, 2013. doi: 10.1186/1471-2288-13-35.

[18] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[19] Yishui Li, Runduo Liu, Jie Liu, Haibin Luo, Chengkun Wu, and Zhe Li. An Open Source Graph-Based Weighted Cycle Closure Method for Relative Binding Free Energy Calculations. *Journal of Chemical Information and Modeling*, 63(2):561–570, 2023. doi: 10.1021/acs.jcim.2c01076.

[20] Nathan M. Lim, Lingle Wang, Robert Abel, and David L. Mobley. Sensitivity in binding free energies due to protein reorganization. *J. Chem. Theory Comput.*, 12(9):4620–4631, 2016. doi: 10.1021/acs.jctc.6b00532.

[21] Guobing Lu and A. E. Ades. Assessing Evidence Inconsistency in Mixed Treatment Comparisons. *Journal of the American Statistical Association*, 101(474):447–459, 2006. doi: 10.1198/016214505000001302.

[22] Antonia S.J.S. Mey, Bryce K. Allen, Hannah E. Bruce Macdonald, John D. Chodera, David F. Hahn, Maximilian Kuhn, Julien Michel, David L. Mobley, Levi N. Naden, Samarjeet Prasad, Andrea Rizzi, Jenke Scheen, Michael R. Shirts, Gary Tresadern, and Haufeng Xu. Best Practices for Alchemical Free Energy Calculations [Article v1.0]. *Living Journal of Computational Molecular Science*, 2(1):18378, 2020. doi: 10.33011/livecoms.2.1.18378.

[23] D.A. Nix and A.S. Weigend. Estimating the mean and variance of the target probability distribution. In *Proceedings of 1994 IEEE International Conference on Neural Networks (ICNN'94)*, pages 55–60 vol.1, 1994. doi: 10.1109/ICNN.1994.374138.

[24] Gregory A. Ross, Chao Lu, Guido Scarabelli, Steven K. Albanese, Evelyne Houang, Robert Abel, Edward D. Harder, and Lingle Wang. The maximal and current accuracy of rigorous protein-ligand binding free energy calculations. *Commun. Chem.*, 6(1):222, 2023. doi: 10.1038/s42004-023-01019-9.

[25] Christina E. M. Schindler, Hannah Baumann, Andreas Blum, Dietrich Böse, Hans-Peter Buchstaller, Lars Burgdorf, Daniel Cappel, Eugene Chekler, Paul Czodrowski, Dieter Dorsch, Merveille K. I. Eguida, Bruce Follows, Thomas Fuchß, Ulrich Grädler, Jakub Gunera, Theresa Johnson, Catherine Jorand Lebrun, Srinivasa Karra, Markus Klein, Tim Knehans, Lisa Koetzner, Mireille Krier, Matthias Leiendecker, Birgitta Leuthner, Liwei Li, Igor Mochalkin, Djordje Musil, Constantin Neagu, Friedrich Rippmann, Kai Schiemann, Robert Schulz, Thomas Steinbrecher, Eva-Maria Tanzer, Andrea Unzue Lopez, Ariele Viacava Follis, Ansgar Wegener, and Daniel Kuhn. Large-Scale Assessment of Binding Free Energy Calculations in Active Drug Discovery Projects. *Journal of Chemical Information and Modeling*, 60:5457–5474, 2020. doi: 10.1021/acs.jcim.0c00900.

[26] Michael R. Shirts and John D. Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of Chemical Physics*, 129(12):124105, 2008. doi: 10.1063/1.2978177.

[27] Daniel A. Spielman and Nikhil Srivastava. Graph Sparsification by Effective Resistances. *SIAM Journal on Computing*, 40(6):1913–1926, 2011. doi: 10.1137/080734029.

[28] Alexander D. Wade, Agastya P. Bhati, Shunzhou Wan, and Peter V. Coveney. Alchemical Free Energy Estimators and Molecular Dynamics Engines: Accuracy, Precision, and Reproducibility. *Journal of Chemical Theory and Computation*, 18(6):3972–3987, 2022. doi: 10.1021/acs.jctc.2c00114.

[29] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K. Dahlgren, Jeremy Greenwood, Donna L. Romero, Craig Masse, Jennifer L. Knight, Thomas Steinbrecher, Thijs Beuming, Wolfgang Damm, Ed Harder, Woody Sherman, Mark Brewer, Ron Wester, Mark Murcko, Leah Frye, Ramy Farid, Teng Lin, David L. Mobley, William L. Jorgensen, Bruce J. Berne, Richard A. Friesner, and Robert Abel. Accurate and Reliable Prediction of Relative Ligand Binding Potency in Prospective Drug Discovery by Way of a Modern Free-Energy Calculation Protocol and Force Field. *Journal of the American Chemical Society*, 137:2695–2703, 2015. doi: 10.1021/ja512751q.

[30] Halbert White. Maximum Likelihood Estimation of Misspecified Models. *Econometrica*, 50:1, 1982. doi: 10.2307/1912526.

[31] Huafeng Xu. Optimal Measurement Network of Pairwise Differences. *Journal of Chemical Information and Modeling*, 59(11):4720–4728, 2019. doi: 10.1021/acs.jcim.9b00528.

[32] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002. doi: 10.1145/775047.775151.

[33] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kiziloren, Anna Gaulton, A. Patricia Bento, Melissa F. Adasme, Peter Monecke, Gregory A. Landrum, and Andrew R. Leach. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.